NOT MEASUREMENT SENSITIVE

MIL-T-62518A
6 November 1991
SUPERSEDING
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MILITARY SPECIFICATION

TRUCKS, MAINTENANCE: WITH 9 TO 37 METERS (30 TO 120 FOOT) MULTI-POSITION AERIAL PLATFORM, MODIFIED COMMERCIAL

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel and gasoline engine driven, four-wheel, two-rear-wheel drive; four-wheel, four-wheel drive; and six-wheel, four-wheel-drive trucks having minimum gross vehicle weights (GVW) of 6800 kilograms (kg) (15,000) to 22 700 kg (50,000 pounds), equipped with maintenance bodies and 9 to 37 meters (m) (30 to 120 foot) multi-position, articulating, boom type aerial platforms. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. The vehicle shall be one of the types, classes and sizes shown in table I, as specified (see 6.2):

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2320

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TABLE I. Vehicle classification.

	Type	Nomenclature	Class	Size
*	I	Articulating boom type aerial platform with limited angle travel of not less than 125 degrees between booms, one, one man basket, size 30 and 35; one,	A	30
			B	30, 35
			C	40, 45
			D	50
*		two man basket, all others.	E	60, 65
*			F	75
			G	85
			H	90, 100
			J	120 (See 6.6)
*	II	Articulating boom type aerial platform with over-center travel between booms (200 degrees minimum), one, one man basket, size 30 and 35; two, one man baskets, all others	A	30, 35
			B	30, 35
			C	35, 40
			D	45, 50

1.2.1 Class. The class of the vehicle shall be determined by the minimum GVW as specified in table II:

TABLE II. GVW ratings.

	GVW rating class		
	Vehicle	kg	(pounds)
	A	6800	(15,000)
	B	8600	(19,000)
	C	10 900	(24,000)
	D	12 700	(28,000)
	E	14 500	(32,000)
*	F	20 900	(46,000)
*	G	21 800	(48,000)
*	H	22 700	(50,000)
*	J	23 600	(52,000)

1.2.2 Size. The size of the vehicle shall be determined by the maximum height the aerial platform can attain, measured from ground level to the platform floor. Sizes shall be the minimum sizes specified in table III. Vehicles furnished in a size actually larger than the minimum specified by the procuring activity shall be furnished in the vehicle class corresponding to that larger size, as indicated in table I.

TABLE III. Sizes.

	Ground to platform floor height	
Size	meters	(feet)
30	9.1	(30)
35	10.7	(35)
40	12.2	(40)

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TABLE III. Sizes. (continued)

Size	Ground to platform floor height	
	meters	(feet)
45	13.7	(45)
50	15.2	(50)
60	18.3	(60)
65	19.8	(65)
75	22.3	(75)
85	25.9	(85)
90	27.4	(90)
100	30.5	(100)
120	(see 6.6)	--

1.2.3 Model. The model of the vehicle shall be determined by the number of wheels and driving wheels. The model available for each class of vehicle shown in table IV shall be as specified (see 6.2):

TABLE IV. Models.

Class	Model	Nomenclature
A, B, C, D, E	4x2	4-wheel, 2-rear wheel drive
	4x4	4-wheel, 4-wheel drive
F, G, H, J	6x4	6-wheel, 4-rear wheel drive

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage: Vehicular, Ignition, Lighting and Starting

* VV-F-800 - Fuel Oil, Diesel.

MILITARY

* MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.

* MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.

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	MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative.
*	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
	MIL-H-83282	- Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft, NATO Code Number H-537.

STANDARDS

FEDERAL

FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.
FED-STD-595	- Colors.

MILITARY

MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MS 51118	- Pintle Assembly, Towing - 40,000 Lbs. Capacity Manual Release.
MS 51317	- Light, Warning, Vehicular, Rotating, D.C.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

* DEPARTMENT OF DEFENSE

Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Occupational Safety and Health Standards.
1926.556 - Aerial Lifts.

(Application for copies of OSHA publications should reference the Code of Federal Regulations, Title 29, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are indicated as DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government document which is current on the date of the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
ANSI A92.2 - American National Standard for Vehicle Mounted Elevating and Rotating Aerial Devices.

(Application for copies of ANSI publications should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM A 53 - Pipe, Steel, Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless, for Ordinary Uses (DoD adopted).
ASTM B 117 - Standard Method of Salt Spray (Fog) Testing (DoD adopted).
ASTM D 4985 - Standard Specification for Low Silicate Ethylene Glycol Base Engine Coolant for Heavy Duty Engines Requiring an Initial Charge of Supplemental Coolant Additive.

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(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70	- National Electric Code.

(Application for copies of NFPA publications should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SAE, INC.

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD adopted).
J537	- Storage Batteries.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
J589	- Turn Signal Switch.
J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J683	- Tire Chain Clearance - Trucks, Buses, and Combinations of Vehicles (DoD adopted).
J688	- Truck Ability Prediction Procedure (DoD adopted).
J704	- Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs (DoD adopted).
J706	- Rating of Winches.
J844	- Nonmetallic Air Brake System Tubing (DoD adopted).
J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connections.
J1292	- Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring.
J1349	- Engine Power Test Code - Spark Ignition and Diesel.

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(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book

* (Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.15, the vehicles, components and accessories to be delivered under the contract as further identified and qualified in the product experience clause of this solicitation shall be standard or optional items sold commercially to the public which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.15, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data book. Special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. The exterior color shall be gloss yellow, matching color chip No. 13538 of FED-STD-595. Extruded, molded or spiral wound booms and baskets need not be painted.

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* 3.1.1.2 Marking and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished.

* 3.1.1.2.1 Warning plates. Warning plates shall be furnished and mounted in a conspicuous location near the lower and upper controls and in the driver's compartment. Plates shall conform to composition A (class l or 2) or composition C of MIL-P-514. At least the following information shall appear on each of the three plates:

WARNING

(a) AERIAL PLATFORM AND OUTRIGGERS MUST BE IN STOWED POSITION WHILE VEHICLE IS BEING MOVED.
(b) RATED PLATFORM LOAD, HYDRAULIC OUTRIGGERS DOWN , NOT IN EXCESS OF _____ POUNDS.
(c) INSPECT PLATFORM LEVELING SYSTEM.
(d) MAXIMUM VOLTAGE BOOM AND PLATFORM WILL WITHSTAND _____ VOLTS AC.
(e) OVERALL TRAVEL HEIGHT WHEN BOOMS ARE IN STOWED POSITION _____ .
(f) VEHICLE SHALL NOT BE OPERATED IN WINDS IN EXCESS OF 30 MPH.
(g) BASKET LINER(S) MUST BE INSTALLED WHEN OPERATING ON OR NEAR ELECTRICAL LINES.
(h) DO NOT PERMIT DIRECT CONTACT OF BASKET TO GROUND WHEN OPERATING ON OR NEAR ELECTRICAL DISTRIBUTION LINES.
(i) MAXIMUM LONGITUDINAL INCLINE _____ DEGREES.
(j) MAXIMUM TRANSVERSE SLOPE _____ DEGREES.
(k) BOOMS AND BASKET(S) MUST BE KEPT CLEAN AND FREE OF DAMPNESS.
(l) DO NOT DRILL HOLES IN BOOM OR BASKET TO DRAIN WATER, DUMP OUT WATER
(m) USE ONLY RECOMMENDED TYPE OF HYDRAULIC FLUID.
(n) WIRE BRAID HYDRAULIC HOSES MUST NOT BE USED ON OR NEAR ELECTRICAL LINES.

* 3.1.1.2.2 Instruction plates. An instruction plate showing a schematic of all valves, valve controls and equipment, and complete warnings and instructions for operating the aerial equipment shall be fastened on the inside of a curbside vertical cabinet door. Each control lever shall be marked with an instruction plate depicting the function of the control. Individual markings shall be adjacent to each control. Plates shall conform to composition A (class l or 2) or composition C of MIL-P-514.

3.1.1.2.3 Warning signs. Warning signs or labels, as recommended by the Manufacturers of Aerial Devices and Digger Derricks Council, shall be furnished and attached.

3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

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3.1.1.4 Drain plugs. All drain plugs installed in manual transmissions, transfer case and rear axles shall be of the permanent magnet type.

* 3.1.1.5 Towing devices. Not less than two hooks, loops, or pins, or a single, center mounted towing eye, for towing the vehicle, shall be furnished on the front of the vehicle. All towing devices shall be frame rail mounted or reinforced back to each frame rail.

* 3.1.1.6 Trailer towing package. When specified (see 6.2), a trailer towing package consisting of a pintle, safety chain attachment devices, a lighting receptacle and associated reinforcements and wiring shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS 51118-1. The pintle shall be installed on the chassis frame with reinforcements to transfer a vertical tongue load of not less than 1815 kg (4,000 pounds) and a longitudinal drawbar load of not less than 178 kilonewtons (kN) (40,000 pounds) directly to the chassis rails. The rearmost portion of the pintle shall be forward but not more than four inches forward of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength of not less than 8150 kg (18,000 pounds). The attachment devices shall be capable of accommodating a standard grab hook [(116 mm (4-9/16 inches) wide, 30 mm (1-3/16 inches) thick, 19.8 mm (25/32-inch) throat width)] for a 16 mm (5/8-inch) chain. The lighting receptacle, conforming to SAE J560 with its conductors connected and color coded as specified therein, shall be mounted in a readily accessible location near the pintle.

3.1.1.7 Trailer lighting cable. A coiled electrical trailer cable conforming to SAE J1067 shall be furnished. The cable shall be not less than 2800 mm (110 inches) long when fully extended. Both ends of the cable shall be equipped with a seven-circuit round plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the vehicle tool compartment.

3.1.1.8 Electric trailer brake controller. When specified (see 6.2), the vehicle shall be provided with an electric trailer brake controller. The controller shall be installed in the truck cab and wired through the SAE J560 receptacle. The controller shall operate on the 12-volt electrical system of the truck and shall include load control, hand lever and accessories.

* 3.1.1.9 Rear wheel splash and stone throw protection. Mud flaps shall be installed to the rear of the rear wheels. A metal strip, not less than 3.2 millimeters (mm) (0.125 inch) thick and not less than 25 mm (one inch) wide, extending the entire width of the mud flap, shall be installed to prevent bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 25 mm (1 inch) high by 32 mm (1.25 inch) wide by 2.4 mm (0.094 inch) thick shall be furnished at each bolt. Rear wheels shall be provided with splash and atone throw protection in accordance with SAE J682.

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3.1.1.10 Drive hubs. When specified for model 4x4, class A and B vehicles (see 6.2), manually operated drive hubs for disengaging the front wheels from the drive train shall be provided, except for vehicles equipped with interaxle compensating devices between front and rear axles (see 3.4.5.1.2), or air brake equipped chassis.

3.1.1.11 Silicone brake fluid. When specified for class A and B vehicles (see 6.2), and if available as manufacturer's standard or optional brake fluid, silicone brake fluid conforming to MIL-B-46176 shall be provided in the hydraulic brake system. A tag on the master cylinder or a permanent decal near the master cylinder shall be furnished stating "CAUTION: USE SILICONE BRAKE FLUID ONLY, MIL-B-46176."

* 3.1.1.12 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

* 3.1.1.13 4x4 conversion. The chassis manufacturer's standard 4x2 truck chassis may be modified to provide all wheel drive conforming to the requirements specified herein if:

(a) The conversion axle manufacturer's engineering department specifically approves and certifies that all such modifications meet the design requirements and standards of the conversion axle manufacturer. Certification shall be based on both design analysis and proving ground test reports, which shall be made available to the engineering and quality assurance offices of the procuring activity.
(b) The chassis manufacturer's front axle before conversion has the same load rating as the conversion axle to be installed.
(c) Components used in the all wheel drive conversion are of current production.
(d) Components used in the all wheel drive conversion are approved for the conversion application by the component manufacturers.
(e) The converted vehicle is certified to conform to Federal Motor Vehicle Safety Standard, No. 121, by the intermediate or final manufacturer.

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(f) Replacement headlights, if required to be added, shall meet the height requirement of not less than 560 mm (22 inches) and not more than 1370 (54 inches), measured above the road surface, in conformance with Federal Motor Vehicle Safety Standard No. 108. Replacement headlights shall be equivalent in mounting, protection, and range and precision of adjustment to the chassis manufacturer's original standard headlights.
(g) Unused headlight cavities are covered in a neat workmanlike manner, treated and painted to match the chassis cab color with treatment and painting equivalent to the chassis cab manufacturer's process for the remainder of the chassis cab. Cavities and their covers shall be rustproofed in accordance with 3.1.1.2.
(h) Complete installation drawings are available to the procuring activity.

* 3.1.1.14 Brushguards. Model 4x4 vehicles shall be equipped with a radiator and headlamp brushguard. When the headlamps are recessed into and protected by the front bumper, a head lamp brushguard is not required.

* 3.1.1.15 Spark arrester. As specified herein (see 3.4.4.1), the vehicle shall be furnished with an exhaust system spark arrester.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with the EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.2.3 Sound level. The cab interior sound level shall not exceed 84 db(a) when measured in accordance with Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior drive-by sound level shall conform to EPA Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

* 3.2.4 Safety standards. The vehicle and rotating aerial device, the manufacturer, the other contributing manufactuer's, dealers (when applicable) and installers (when applicable) shall comply with ANSI A92.2. The vehicle shall comply with all OSHA requirements, including OSHA 1926.556.

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3.2.5 Curb weight. The curb weight shall include the weight of the chassis and the cab, with all attachments, accessories and equipment; body and aerial platform assembly in stowed position; and a full complement of fuel, lubricants, and coolant.

3.2.6 Gross vehicle weight. Gross vehicle weight (GVW) shall consist of curb weight, operator (weight computed at 80 kg (175 pounds)), and a payload uniformly distributed throughout the length of the vehicle body, to provide not less than the specified GVW. Vehicles with a crew (four-door) cab shall have GVW computed with 240 kg (525 pounds) of payload in the rear seat.

3.2.7 Weight distribution. The axle capacity and the capacity of the suspension and tires per axle shall be not less than as specified in table V for the specified type, class and size of vehicle.

TABLE V. Suspension, axle, and tire capacities.

	Type	Class	Size	Front axle	Rear axle	GVW
				kg (pounds)	kg (pounds)	kg (pounds)
	I	A	30	2270 (5,000)	4994 (11,000)	6800 (15,000)
		B	30 and 35	3180 (7,000)	6350 (14,000)	8600 (19,000)
		C	40 and 45	4085 (9,000)	8150 (18,000)	10 900 (24,000)
*		D	50	4994 (11,000)	8980 (19,800)	12 700 (28,000)
		E	60	5450 (12,000)	9070 (20,000)	14 500 (32,000)
*		F	65 and 75	5450 (12,000)	15 400 (34,000)	20 900 (46,000)
*		G	85	6350 (14,000)	15 400 (34,000)	21 800 (48,000)
*		H	90 and 100	7250 (16,000)	15 400 (34,000)	22 700 (50,000)
*		J	120	8150 (18,000)	15 400 (34,000)	23 600 (52,000)
	II	A	30 and 35	2270 (5,000)	4994 (11,000)	6800 (15,000)
		B	30 and 35	3180 (7,000)	6350 (14,000)	8600 (19,000)
		C	35	4085 (9,000)	8150 (18,000)	10 900 (24,000)
		C	40	4670 (10,300)	8150 (18,000)	10 900 (24,000)
*		D	45 and 50	4994 (11,000)	8980 (19,000)	12 700 (28,000)

3.2.8 Ratings. The vehicle ratings shall be manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity. Minimum GVW ratings shall be not less than specified in table II for the specified class of the vehicle. The ratings shall be established by the manufacturer of the equipment being verified.

3.2.9 Cab-to-axle. The cab-to-axle or cab-to-trunnion dimension shall be not less than as specified in table VI. The dimension shall be measured from the back of the cab or crew cab to the center of the rear axle or trunnion, as applicable.

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TABLE VI. Minimum cab-to-axle dimensions.

Class	Type	Size	Minimum dimension	
			mm	(inches)
A	All	All	1520	(60)
B	I	All	2105	(83)
C	I	40	2105	(83)
C	I	45	2540	(100)
D	All	All	2540	(100)
E	I	60	3810	(150)
F	I	All	3500	(138)
G	I	All	3500	(138)
H	I	All	3500	(138)
J	I	All	3500	(138)
B	II	All	2540	(100)
C	II	All	2540	(100)

* 3.2.10 Overall width and height. The overall width of the vehicle, exclusive of tires and safety related items such as mirrors, lights and reflectors, shall be not more than 2440 mm (96 inches). The width over the tires shall be not more than 2540 mm (100 inches). The overall height of the vehicle, with the booms stowed for over road use, shall be not greater than as specified in table VII.

TABLE VII. Maximum overall height.

Size	Model	Maximum overall height	
		mm	(feet)
30, 35	4x2, 4x4	3810	(12.5)
40 to 60	4x2	3810	(12.5)
40 to 60	4x4	4110	(13.5)
65, 75	6x4	3810	(12.5)
85 and up	6x4	4110	(13.5)

3.2.11 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.12 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

3.3 Performance. Performance requirements for model 4x4 vehicles shall be met with the front wheel drive disengaged and with front wheel hubs (when furnished) unlocked except that vehicles equipped with interaxle compensating devices shall meet performance with the front wheel drive engaged.

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3.3.1 Speeds. High and low speed requirements shall be met with the vehicle loaded to specified GVW.

3.3.1.1 High speed gradeability. The vehicle shall ascend continuous grades specified in table VIII at 80 kilometers per hour (km/h) (50 miles per hour (mph)). Gradeability requirements shall be met with the main transmission in direct drive and, when a multispeed axle or auxiliary transmission in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

* TABLE VIII. Gradeability.

Vehicle class	Percent grade
A	3.0
B	3.0
C	2.5
D	2.0
E	1.5
F	1.1
G	1.0
H	1.0
J	1.0

3.3.1.2 Low speed. Low speed for vehicles with a manual transmission shall be calculated with the engine operating at not less than 35 percent of the recommended governed speed, and shall provide a vehicle speed of not more than 6.4 km/h (4 mph).

* 3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 84 km/h (52 mph) for model 4x4 vehicles and not less than 93 km/h (58 mph) for model 4x2 and 6x4 vehicles. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (km/h)} \quad \frac{\text{Governed speed (rpm) x 1.609}}{\text{Total gear reduction x tire factor (see 6.3).}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

* 3.3.3 Upsetting stability. With 1-1/2 times the specified load (see 3.6.5) placed in or suspended from the aerial basket(s), with the vehicle otherwise unloaded, and with all the outriggers extended or with the torsion bars on sizes 30 and 35, the articulated booms shall be capable of placing and holding the loaded aerial basket(s) in any attainable position of the

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equipment without causing the vehicle to become unstable. The vehicle shall be deemed unstable with 1-1/2 times the specified load when anyone of the vehicle wheels or outriggers lifts off the ground, ANSI A92.2 not withstanding. The vehicle shall also be deemed unstable if, with twice the specified load (see 3.6.5), the inner tire of either dual rear wheel lifts off the ground. The stability tests shall not cause any permanent deformation beyond the elastic limit of any component of the vehicle.

* 3.3.3.1 Stability on slopes. With one and one third times the specified load (see 3.6.5) placed in or suspended from the basket(s), with the vehicle otherwise unloaded, with the vehicle on a 5 degree slope, with the outriggers extended, without the vehicle being leveled, (ANSI A92.2 not withstanding) facing upward, then downward, then at right angles to the 5 degree slope, the articulated booms shall be capable of placing and holding the loaded aerial basket(s) in any attainable position of the equipment without causing the vehicle to become unstable. The vehicle shall be deemed unstable with one and one third times the specified load when anyone of the vehicle's tires or outriggers lifts off the ground, ANSI A92.2 not withstanding.

3.3.4 Wind resistance. With the specified load (see 3.6.5) in the basket(s), with the vehicle otherwise unloaded, and in any extended position, the vehicle shall be capable of withstanding a wind load of 335 N/m² (7 pounds per square foot) without upsetting, overturning, or permanent deformation of any component. The maximum area exposure for wind load shall be calculated with the body, booms and basket(s) in the most critical position that would induce tipping. The resisting moment shall be greater than the wind load when the load is applied on the surface of the body, boom and basket(s). The contractor shall perform calculations, or shall have available a certified test report for the vehicle or a vehicle of the same make, type, class, size and model.

3.3.5 Height. The booms shall be capable of raising the aerial basket(s) with rated load (see 3.6.5), measured from ground level to the aerial basket(s) floor, to the minimum height for the size of vehicle specified in table III.

* 3.3.6 Horizontal reach. With the lower boom in its most horizontal position for type II, and its vertical or most nearly vertical position for type I, the boom assembly with the upper boom horizontal shall provide a minimum horizontal reach conforming to table IX, as specified for the type and size of the vehicle, measured from the front edge of the aerial basket(s) to the axis of the turntable. When side hung basket(s) are furnished on type I, the horizontal reach specified in table IX may be reduced not more than 535 mm (1-3/4 feet). The booms shall be capable of lowering the loaded aerial platform to within 1220 mm (4 feet) of the ground, measured from the ground to the side of the basket nearest the ground, when the basket has been tilted over as to remove a disabled operator. The tilt control, before and after the basket is tilted, shall be not

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more than 1520 mm (5 feet) above ground level. The booms of type II vehicles shall be capable of lowering the loaded aerial platform to ground level.

TABLE IX. Horizontal reach.

Type	Size	Horizontal reach	
		mm	(feet)
I	30	5.5	(18)
	35	6.4	(21)
	40	6.7	(22)
	45	7.0	(23)
	50	8.8	(29)
	60	10.3	(34)
	65	10.3	(34)
	75	12.5	(41)
	85	12.5	(41)
	90	12.5	(41)
	100	13.4	(44)
	120	13.4	(44)
II	30	7.3	(24)
	35	8.2	(27)
	40	9.5	(31)
	45	11.3	(37)
	50	12.8	(42)

3.3.6.1 Extended horizontal reach. When specified (see 6.2), the boom assembly shall provide the extended horizontal reach specified in table X in lieu of the reach specified in table IX.

Table X. Extended horizontal reach (feet).

Size	Extended horizontal reach	
	meters	(feet)
75	13.0	(42.5)
85	15.2	(50.0)
90	15.8	(52.0)
100	16.5	(54.0)

* 3.3.7 Rotation (boom assembly). The boom assembly shall be capable of continuous rotation in either direction in the horizontal plane. The turntable motor shall rotate the loaded basket(s) through an arc of 360 degrees in not less than 60 seconds and not more than 230 seconds. The turntable rotating device shall remain in a fixed position when the hydraulic motor is de-energized. The turntable shall be capable of smoothly rotating the aerial platform assembly up and down a 5 percent grade when loaded to the rated working capacity, and producing the maximum attainable tipping moment.

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3.3.8 Position stability (aerial platform). When the vehicle is on a 5 percent grade, without leveling the vehicle, the elevating and rotating mechanism shall hold the loaded aerial platform (see 3.6.5) in any extended position within 25 mm/hr (1 inch per hour) horizontal creep and 50 mm/hr (2 inches per hour) vertical settling. The creep and settling shall be measured at the front of the aerial platform with hydraulic fluid stabilized at ambient air temperature.

3.4 Chassis components.

3.4.1 Engine. The engine furnished for the specified vehicle class shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than six cylinders. Engine net power figures used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable.

* 3.4.1.2 Gasoline engine. When specified for model 4x2 (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at an engine speed not more than the manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel, when used in accordance with the operator's manual. Engine net power figures used in performance prediction calculations shall be determined in accordance with SAE J1349. Unless otherwise required by foreign laws, vehicles for overseas shall be capable of accepting and operating on leaded gasoline.

3.4.1.3 Oil filter. A full flow type oil filter shall be furnished.

3.4.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

* 3.4.1.5 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at an altitude of 3050 meters (m) (10,000 feet) above sea level or in an ambient air temperature of not

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less than 52° Celsius (ºC) (125° Fahrenheit (ºF)). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than 1.89 liters (L) (two-quart) capacity. For cooling system servicing, see 3.4.23.1.

3.4.1.6 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.7 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

* 3.4.1.8 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, and fuel warmers (for diesel engine driven vehicles only) shall be provided. Heaters shall operate on 120-volt alternating current (ac), and shall be wired through a junction block, including a fuse or circuit breaker, to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and grounding vehicle. The receptacle shall be located on the streetside of the vehicle, as near the cab door as possible. A three-wire connecting cable, not less than 7600 mm (25 feet) long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard weatherproof, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to -51°C (-60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters and fuel warmers shall be furnished as follows:

(a) A coolant heater, 1500-watt (W) minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. An engine thermostat with an operating range of 77°C (170°F) to 90°C (195°F) shall be installed.

* (b) An oil pan heater of the permanent external surface mount, permanent in-pan mount, or immersion type that meets the following requirements, shall be installed:

* (1) Not less than 16 W/L (15 watts per quart) heating capacity

* (2) Not more than 2.8 watts per square centimeter (W/cm²) (18 watts per square inch) heating capacity

(3) Thermal balance design or thermostat control providing for uninterrupted operation

(4) Provision for mounting below minimum service oil level.

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* (c) An in-line fuel warmer or preheater unit shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The unit shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40°C (-40°F) to an outlet temperature of -13°C (+9°F), with a fuel flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit. The unit shall not cause heating of the fuel above 27°C (80°F) under any possible condition.

* (d) An in-tank fuel warmer or preheater unit shall be provided on diesel engine driven vehicles. The unit shall use engine coolant to transfer heat to the fuel in one fuel tank. The warmer shall not cause heating of any fuel above 27ºC (80°F) under any possible condition, shall not disable or cause elimination of the fuel gage sending unit and shall not violate 3.2.1 or any Federal Motor Carrier Safety Regulation. A coolant shut off valve shall be included. The units required by 3.4.1.8(c) and (d) may be combined.

* 3.4.1.9 Vernier throttle control. When specified (see 6.2), a manual vernier throttle with quick release shall be furnished, except on electronically controlled engines.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system (diesel engines). For diesel engine driven vehicles, a 12- or 24-volt direct current (dc) starting system, with a 12-volt dc lighting system shall be furnished. Engine starting equipment shall include an ether starting system, glow plug or grid heater. If an ether system is furnished in lieu of a glow plug or grid heater, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment, and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 355 milliliters (mL) (12 fluid ounces) shall be furnished. A reservoir need not be furnished. When specified (see 6.2), the starting motor shall be equipped with a thermostat controlled, automatic resetting circuit breaker that will protect the motor from overcrank heat damage. The circuit breaker shall not limit cranking ability for a startable engine.

* 3.4.2.2 Ignition system (gasoline engines). For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished.

* 3.4.2.3 Alternator. Unless otherwise specified (see 6.2), a minimum 85-ampere alternator shall be furnished. The alternator output with the engine at engine idle speed shall be not less than 40 amperes.

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3.4.2.4 Lighting. All vehicle lights and reflectors shall be in accordance with Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on the vertical surfaces of rub rails (unless recessed and fully protected) or on vehicle bumpers.

* 3.4.2.5 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Turn signal units shall be installed in accordance with SAE J588. Turn signal operating units shall have visible flash indicators.

3.4.2.6 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating and the total cold cranking rating at 0ºF, both measured in accordance with SAE J537, shall be not less than specified in table XI. The batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

* TABLE XI. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel, 200 kilowatts (kW) gross (270 gross horsepower) or less	480	1,875
Diesel, over 200 kW gross (270 gross horsepower)	640	2,500
Gasoline	100	450

3.4.2.7 Radio interference suppression. The vehicle shall be suppressed to limit the electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

* 3.4.3.1 Air cleaner. An air cleaner shall be furnished.

* 3.4.3.2 Fuel tank(s). The vehicle shall be equipped with fuel tank(s) of not less than 76 L (20-gallon) capacity for class A vehicles and not less than 189 L (50-gallon capacity) for all other vehicles (163 L (43 gallons) minimum for other than class A vehicles destined for California)). When more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure an equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve connecting either tank to the engine fuel intake shall be provided and means shall be provided to monitor the fuel level of

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either tank from a single fuel gage; or an equalizing pump shall be used to maintain the same fuel level in both tanks.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. When vertical exhaust mufflers are furnished and are capable of being reached easily by personnel entering or leaving either side of the cab, they shall be provided with a heat shield. Vertical mufflers, when furnished, shall be furnished with a hinged rain cap.

3.4.4.1 Spark arrester. A spark arrester shall be furnished, except on turbocharged engines. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350.

3.4.4.2 Exhaust system safety shield. The exhaust system or the hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

3.4.5 Gear train.

3.4.5.1 Transmission. Unless otherwise specified (see 3.4.5.3), a manually shifted transmission shall be provided on the vehicle. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall be provided with a power takeoff opening conforming, for manual transmissions, to SAE J704. The transmission shall provide for maximum ease of shifting in all speeds.

* 3.4.5.1.1 Speeds. The vehicle shall be provided with not less than the number of forward speeds specified in table XII. When more than five forward speeds are required, a two-speed axle (model 4x2 and 6x4 only), multispeed transmission, auxiliary transmission or two-speed transfer case (model 4x4 only) shall be furnished. Gear ratios in the transmission(s) and axle(s) shall be matched to provide a progressive shifting pattern throughout the complete range, and shall provide the required vehicle performance specified in 3.3.1 through 3.3.1.3.

TABLE XII. Vehicle requirements, minimum number of forward speeds.

Vehicle Model	Vehicle class	No. of forward speeds
4x2	A, B, C	5
4x2	D, E	8
* 4x4	A, B, C, D, E	9
* 6x4	F, G, H, J	13

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* 3.4.5.1.2 Transfer case (model 4x4). Model 4x4 vehicles shall be provided with a transfer case. Unless otherwise specified, either a one- or two-speed transfer case shall be provided. Unless the transfer case is equipped with devices which compensate for differential torque and speeds between the front and rear axles, the transfer case shall provide for driver selection of either two-wheel or four-wheel drive. When interaxle compensating devices are furnished, they shall provide for the positive transfer of power to all the driving axles. When specified (see 6.2), a two-speed transfer case shall be furnished. On two-speed transfer case vehicles, the speedometer shall read accurate vehicle speed with the speed selector on high and in low range.

* 3.4.5.2 Clutch. The clutch shall be the largest capacity clutch offered for the type, class and model of vehicle and engine furnished with clutch torque capacity exceeding maximum delivered engine torque. The clutch lining shall be asbestos free. The clutch on 6x4 models shall be equipped with spring dampening and a greaseable bearing.

* 3.4.5.3 Automatic transmission. When specified (see 6.2), the vehicle shall be provided with an automatic transmission. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios for model 4x2 and 4x4 vehicles, and not less than five forward gear ratios for model 6x4 vehicles. The normal driving range selector position shall provide not less than four forward gear ratios without movement of the selector. A drive gear for the power takeoff shall be provided.

3.4.5.4 Power takeoff. A power takeoff is required. The power takeoff shall be of a rated capacity to operate all auxiliary equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the type of vehicle furnished. The vehicle shall have a frame with reinforcements extending at least from the rear of the front-suspension, rear hanger bracket to beyond the rear spring front hanger bracket or to the bogie trunnion mounting bracket. Reinforcement shall provide sufficient structural strength in the chassis frame to adequately distribute and support the additional loads imposed by the rotating hydraulic aerial platform and the outrigger supports. Reinforcements shall provide

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sufficient structural strength in the chassis frame through increased bending moment to at least equal the loads imposed with the vehicle body loaded to provide the specified GVW.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a total rated capacity per axle at least equal to that specified in 3.2.6. The vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels. Suspension system ratings shall be adequate for the loads imposed during all conditions of operation of the aerial equipment.

* 3.4.9 Axles. Axle ratings shall be at least equal to that specified in 3.2.6. Gear ratio(s) shall provide performance specified in 3.3.1 through 3.3.1.3. Except for the front axle or model 4x4 vehicles, the wheel bearings and axle spindles shall be oil lubricated. The hubcaps, except for driving axles, shall have a window for visual determination of oil level. Provisions for venting or withstanding internal pressure buildup and for replenishing the oil supply shall be provided.

3.4.9.1 Rear bogie. The rear bogie on model 6x4 vehicles shall be of the four-wheel type, complete with axles, springs, torque rods and all other necessary parts. The bogie shall be provided with means permitting differential action between the two axles, and a manually or automatically controlled lockout assuring equal power to each rear axle. The manual lockout control used shall be located in the truck cab. Axle gear ratios shall provide performance specified in 3.3.1 through 3.3.1.3.

3.4.9.2 Two-speed axles. When specified for model 4x2 and 6x4 (see 6.2), a two-speed axle(s) shall be furnished. The axles shall be equipped with electric, vacuum, or air shift and shall be provided with ratios which will permit proper transmission gear splitting. Gear ratios shall provide performance specified in 3.3.1 through 3.3.1.3.

3.4.9.3 Traction control. When specified (see 6.2), the vehicle shall be furnished with a traction control on the rear axle for model 4x2 and 4x4 vehicles and on either rear axle for model 6x4 vehicles. The traction control shall actuate automatically to insure that power is transmitted to the wheels having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.10 Wheels, rims, tires and tubes. Unless otherwise specified, the vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to The Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Except when tube type tires are specified in the procurement documents, multi-piece rims shall not be furnished on front axles rated at 6350 kg (14,000 pounds) or less and on single rear axles rated at 10430 kg (23,000 pounds) or less or

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tandem axles rated at 20 900 (46,000 pounds) or less. Except for class F, G, H and J, rims and disc wheel sizes shall be the same for all wheels on each vehicle. Except for class F, G, H and J, tire size and load range (ply rating) shall be the same for all tires on each vehicle. When specified (see 6.2), vehicles shall be equipped with wide base type tires and wheels for the front and rear axles in lieu of the duals. Wide base wheels shall be interchangeable without the use of an adapter. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.10.1 Tires. Except for class A vehicles, tires shall be steel belted radial, or when specified (see 6.2), bias ply type. Tires for class A vehicles shall be of the bias ply type. Unless otherwise specified, low profile or standard profile tires may be used. When specified (see 6.2), low profile tires shall be furnished. Tires shall be tubeless type with highway tread for model 4x2 and 6x4 vehicles and with non-directional mud and snow tread for model 4x4 vehicles. Tires shall be of a rated capacity at least equal to that specified in 3.2.6. Tires shall conform to Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

* 3.4.10.2 Inner tubes. When tube type tires are specified in procurement documents, inner tubes shall be of the heavy duty type, and shall be of proper size for the tires furnished. Tube type radial tires shall have radial tire inner tubes. Tire flaps shall be provided for tube type tires in accordance with The Tire and Rim Association recommendations.

* 3.4.10.3 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be furnished. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier shall be mounted vertically in the body load space. The carrier design shall enable removal or mounting of the spare wheel assembly using only the tools specified in 3.4.16.1. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim identical to those furnished on the front axle of the vehicle shall be provided. When a spare tire assembly is specified (see 3.4.10.5), a spare wheel or rim shall be furnished.

* 3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the front axle of the vehicle.

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3.4.10.6 Tire chain clearance. Tire chain clearance in accordance with SAE J683 shall be provided. Allowance for spring deflection shall be included.

* 3.4.11 Brakes. The brakes shall be in accordance with Federal Motor Carrier Safety Regulations 393.40 through 393.43, and 393.45 through 393.52. Brake linings shall be of nonasbestos material.

* 3.4.11.1 Power hydraulic type brakes. Unless otherwise specified (see 3.4.11.2 and 3.4.11.4), class A and class B, model 4x2 vehicles shall be equipped with power assisted, hydraulic actuated, four-wheel service brakes.

3.4.11.2 Full air type brakes. Model 4x4 and 6x4 vehicles and model 4x2, class C, D, E vehicles and, when specified (see 6.2), class A and class B, model 4x2 vehicles shall be equipped with full air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air or water-cooled, and having a capacity of not less than 205 l/min (7.25 cubic feet per minute (cfm)) for model 4x2 and 4x4, and not less than 340 L/min (12 cfm) for model 6x4
(b) Air storage reservoir(s); each tank equipped with a draincock and with safety and check valves between the compressor and the last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low air pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
(i) Automatic slack adjusters (model 4x2 and 6x4 only).

* 3.4.11.3.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of purge valve.

* 3.4.11.4 Trailer brake control system. When specified (see 6.2), a trailer brake control system shall be furnished in addition to the components specified in 3.4.11.2. When a trailer brake control system is specified, class A and B vehicles also have an air brake system in lieu of hydraulic brakes. The trailer brake control system shall include:

(a) Identification of emergency and service lines
(b) Coincident control of trailer brakes with prime mover foot control

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(c) Independent hand control for trailer brakes
(d) Towing vehicle protection valve with dash control and automatic breakaway feature
(e) Trailer stoplight control operable with foot brake and with hand control for trailer brakes
(f) Air connectors for trailers with SAE J318 gladhand couplers mounted at the rear of the vehicle, located to interference with trailer
(g) Two SAE J844 coiled air hoses, each not less than 110 inches long when fully extended, with SAE J318 gladhand couplers on both ends. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment
(h) Dummy gladhand couplers with chains or cables
(i) Prime mover only parking brake valve to permit prime mover brakes to be applied while charging the trailer air brake system.

* 3.4.11.5 Brake control for use from a towing vehicle. When specified for vehicles with air brakes (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brake is furnished), additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.12 Cab. Unless otherwise specified, the chassis manufacturer shall furnish any type of his standard or optional non-tilting full width cab. When specified (see 6.2), a cab with a forward tilting hood and fender assembly, including tilting and locking mechanism, shall be furnished. Tilting shall not interfere with present or future installations of additional equipment, such as a snowplow or a front mounted winch. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, with both doors equipped with an external, key-operated locks. Drip rails shall be installed above the cab doors. When the front tires extend beyond the cab fenders, rubber fender extensions extending at least to the outside of the tire tread shall be furnished.

* 3.4.12.1 Cab interior. Unless otherwise specified, the cab shall have an upholstered, full-width, adjustable seat and back or individual, adjustable driver's seat and individual passenger seat. When specified (see 6.2), an individual, adjustable driver's seat and an individual passenger seat shall be provided. When specified for vehicles with air brakes (see 6.2), the driver’s seat shall be the manufacturer’s standard air ride suspension type. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White

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upholstery shall not be furnished. Interior lighting shall be provided. Two pairs of seat belts shall be installed. Outboard seats shall have combination pelvic and upper torso restraint seat belts.

* 3.4.12.2 Crew cab. When specified (see 6.2), a four-door, full width crew cab shall be furnished in lieu of the standard full width cab. The cab shall be equipped with two upholstered full width seats and backs. The front seat shall be adjustable. Three pairs of seat belts shall be installed for both the front and rear seats. Outboard front seats shall have combination pelvic and upper torso restraint seat belts. The cab doors shall be equipped with locks operable from inside the cab through mechanical linkage, with both front doors equipped with external key-operated locks. Cab doors shall have windows with crank-operated window regulators. A rear window shall be provided. Interior lighting shall be provided. The cab roof shall be of one-piece construction; or, if welded, the roof shall give the appearance of one-piece, with weld seams being continuous, waterproof, and free of visible bumps or protrusions. Full length drip moldings shall be mounted above the doors. When a crew cab is furnished, the cab-to-axle dimension shall be measured from the back face of the crew cab.

3.4.12.3 Crew cab seat spacing. With the front seat adjusted to the extreme forward position, there shall be not less than 760 mm (30 inches) measured in a horizontal plane, between the front of the rear seat-back and the rear of the front seat-back. A kick-space height of not less than 70 mm (2.75 inches) shall be maintained between the floor and the rear of the front seat in all positions of adjustment. Leg room and space forward of the front seat shall be equivalent to that provided ahead of the seat in a two-door standard cab.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type operated by air or electric motor(s). When specified (see 6.2), and if electric motor wipers are furnished, intermittent wipers shall be furnished.

3.4.15 Bumpers. Unless the front bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided, except when a front mounted winch is specified, a pipe type front bumper may be furnished (see 3.4.22.2). The rear bumper shall conform to 3.5.1. Rear end protection shall be provided and shall be in accordance with Federal Motor Carrier Safety Regulation 393.86.

* 3.4.16 Tool stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains (for outside tires on duals only) and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools may

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be furnished inside the cab or body cabinets. When stowage space for these tools is located outside the cab or cabinets, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly and shall include at least a hydraulic jack, jack handle, and wheelnut wrench. The jack shall be of such closed height as to permit its location under an axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of the wheel and tire assembly.

3.4.17 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to the floor and to defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted.

3.4.18.1 Accessories and equipment. The chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or red indicator warning light
* (e) Engine coolant temperature gage or high coolant temperature or low coolant level red indicator warning light
* (f) High coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer
(h) Tachometer
(i) Ash receptacle
(j) Dual sunvisors
(k) Driver’s compartment ventilator other than window.

* 3.4.18.2 Rearview mirrors. Outside rearview mirrors shall be mounted on both sides of the cab. The mirrors shall have flat and convex areas. The flat portion shall have not less than 320 cm² (50 square inches) of reflective area. The convex portion shall have not less than 155 cm² (24 square inches) of reflective area and a radius of curvature of not less than 510 mm

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(20 inches). The convex portion shall be attached to the lower mirror supporting arm and shall not interfere with use of the flat mirror. The mirrors shall have not less than two supporting arms. When specified (see 6.2), the curbside flat mirror shall be of the motorized type, with remote control. The mirror motor shall provide not less than 60 degrees horizontal rotational viewing range. When specified (see 6.2), the flat mirrors shall be electrically heated. Mirror remote and heating controls shall be within reach of the seated driver.

* 3.4.19 Horn. The manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, for vehicles with air brakes, an air operated horn shall be furnished.

* 3.4.20 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.4.21 Backup alarm. When specified (see 6.2), the vehicle shall be provided with an audible, pulsating, signaling device (electrical or mechanical) to caution personnel when the vehicle is in a reverse gear operation.

* 3.4.22 Front mounted winch. When specified (see 6.2), the vehicle shall be furnished with a winch assembly, mounted on the front of the vehicle. The winch shall be powered by the aerial platform hydraulic system. The winch shall conform to SAE J706. The winch shall be of the single drum type for hoisting and negotiating purposes and shall have one forward and one reverse speed. Dual winch controls shall be furnished, one inside the cab and one at the front bumper. Cab winch controls shall be so located as to provide no interference with the entry or exit of the driver. The winch shall have a rated single line pull capacity of not less than 9070 kg (20,000 pounds) on the bare drum. The winch shall be equipped with a hydraulic relief valve set to restrict the line pull capacity to 9980 kg (22,000 pounds). The winch line speed on the bare drum shall be between 4570 (15) and 9140 mm (30 feet per minute) at an engine speed equal to 50 percent of engine governed speed. The winch shall be wound with not less than 56 m (185 feet) of 16 mm (0.625 inch) diameter, preformed, 6x37, improved plow steel, independent wire rope core (IWRC) wire rope and shall be equipped with an end chain and hook. An integral, adjustable automatic safety brake shall be provided. The winch shall be equipped with a 4-way roller guide assembly. The winch shall be equipped with a shaft extension on the curbside with a capstan drive extension (AT&T type).

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3.4.22.1 Winch drum guard. A winch drum guard shall be furnished. The guard shall confine the cable to the area between the drum flanges. The guard shall consist of not less than 6.4 mm (0.25 inch) vertical side plates, conforming to the outside radius of the drum flanges. Six bars, 9.5 mm (0.375 inch) by 32 mm (1.25 inches), shall be welded to the vertical side plates. Three bars shall be located on the top and spaced equally on the top radius, and three bars shall be located on the bottom and spaced equally on the bottom radius. The vertical distance between the vertical side plates and the flanges shall be not more than one-half the specified cable diameter.

3.4.22.2 Front bumper. When a front mounted winch is specified, the bumper shall be mounted forward of the winch. The open area on either side of the winch shall be covered with a combination step plate and gravel guard. The bumper shall be either a channel or pipe type. When a pipe type front bumper is furnished, the nominal diameter shall be not less than 75 mm (3 inches), and shall have a wall thickness of not less than that specified in Schedule 40 of ASTM A 53. The pipe type front bumper shall have half-round ball ends.

* 3.4.22.3 Combination step plate and gravel guard. When a front mounted winch is specified, a combination step plate and gravel guard shall be furnished. The step plate shall be fabricated of not less than 14 gage (1.897 mm) (0.0747 inch) steel tread plate exclusive of the projections. The step plate shall be secured to the front bumper and shall be not less than 6.4 mm (0.25 inch) or not more than 9.5 mm (0.375 inch) from the chassis sheet metal. Rubber flaps may be employed to achieve these dimensions and shall be furnished when required by tilting hood/fender vehicles for unrestricted tilting. The step plate shall be capable of supporting 1460 kg/m2 (300 pounds per square foot). The step plate shall deflect not more than 3.2 mm (0.125 inch) under the loads imposed.

3.4.22.4 Winch reel. A reel for payout and take-up shall be provided when a front mounted winch is furnished. The reel shall be designed to operate from the end of the extended winch shaft. Reel yokes shall be controlled by a handle at the end of the reel shaft. The spindle shall be fitted with a bayonet type socket for attachment to the shaft. Reel spiders and yokes shall be made from heat treated aluminum. The inside diameter of the coil shall be approximately 510 mm (20 inches), the maximum outside diameter approximately 710 mm (28 inches), and the thickness or width of the coil approximately 150 (6) to 230 mm (9 inches).

* 3.4.23 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: alinement of lights; adjustment of engine; adjustment of brake system; filling and charging of batteries; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engine, running gear, and mounted equipment with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing

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of cooling system in accordance with 3.4.23.1; and servicing of the windshield washer reservoir with water and appropriate additives. Cable, where used, shall be lubricated with proper lubricants to resist corrosion.

* 3.4.23.1 Engine coolant. The engine coolant shall be a solution of ethylene glycol antifreeze and water or propylene glycol antifreeze and water, in equal parts of antifreeze and water by volume. Ethylene glycol antifreeze shall conform to ASTM D 4985 with not more than 250 parts per million silicates. Propylene glycol antifreeze shall be in the specific formulation approved by the engine and cooling system manufacturers. When specified (see 6.2), the percentage of antifreeze in the cooling system shall be increased to provide protection against freezing down to -54°C (-65°F).

3.5 Body. The vehicle shall be equipped with a line maintenance and construction type body having the minimum dimensions and loading requirements specified in table XIII for the applicable cab-to-axle (see 3.2.8, table VI). The minimum cabinet space length in table XIII, for vehicles with a crew cab, shall be increased not less than 890 mm (35 inches) unless the outriggers are mounted ahead of the cabinet space just behind the crew cab. The body shall include weatherproof cabinets on each side of a center loadspace with a front bulkhead, a hydraulically operated aerial platform assembly, and other equipment specified herein. Cabinet space may be utilized to accommodate hinged type outriggers or access steps. Cabinet areas are cut to provide clearance for A-frame telescopic type outriggers shall be covered with sheet metal to provide a completely enclosed and weatherproof cabinet. Safety tread steel or aluminum plate shall be furnished on the top of the cabinets, the floor and the platform extension.

TABLE XIII. Body dimensions.

Body dimensions	Cab-to-axle				
(minimum)	1520 (60)	2105 (83)	2540 (100)	3505 (138)	3810 (150)
Overall width of body mm (inches)	2160 (85)	2158 (85)	2290 (90)	2290 (90)	2380 (94)
Loadspace width (exclusive of wheelwells mm (inches)	1270 (50)	1270 (50)	1270 (50)	1270 (50)	1270 (50)
Cabinet space (each side including wheelwells): Length mm (inches) without crew cab	2690 (106)	3255 (128)	3680 (144)	5335 (210)	4270 (168)

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TABLE XIII. Body dimensions. – (continued)

Body dimensions	Cab-to-axle				
(minimum)	1520 (60)	2105 (83)	2540 (100)	3505 (138)	3810 (150)
Length mm (inches) with crew cab	1730 (68)	2290 (90)	2715 (107)	4445 (175)	3325 (131)
Height at front mm (inches) exterior	1020 (40)	1070 (42)	1140 (45)	1140 (45)	1140 (45)
Height at rear mm (inches) exterior	965 (38)	1020 (40)	1095 (43)	1095 (43)	1095 (43)
Width mm (inches) exterior	370 (14.5)	380 (15)	380 (15)	380 (15)	380 (15)
Loading (without permanent deformation):					
Floor of usable loadspace, live load on any square meters (square foot) kilograms (pounds)	12 700 (500)	12 700 (500)	12 700 (500)	12 700 (500)	12 700 (500)
Cabinets and shelves, live load on any square meter (square foot) kilometers (pounds)	2540 (100)	2540 (100)	2540 (100)	2540 (100)	2540 (100)

3.5.1 Bumper and platform extension. A combination bumper and body platform extension shall be provided, extending to the rear not less than 150 mm (6 inches) beyond the cabinets. Steel or aluminum safety tread plate shall be provided on top of the platform extension.

3.5.1.1 Stirrup steps. Stirrup steps shall be furnished at the rear on both aides, beneath the combination bumper and platform extension. Grab handles shall be installed to assist personnel in entering the body. Stirrup steps and grab handles shall be furnished on the rear of both rear cabinet compartments for access to the top of the cabinets.

3.5.1.2 Access steps. When the turntable is mounted so that a loadspace length of 910 mm (3 feet) or more is forward of the turntable, access steps and grab handles shall be furnished on the curbside of the body, located between the back of the cab and the turntable. Cabinet space may be reduced on the curbside to accommodate the access steps.

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3.5.1.3 Cab protector. A cab protector to provide protection from falling objects shall be provided. The cab protector framing shall be covered with expanded sheet steel, of not less than 9 gage (3.793 mm) (0.1495 inch) prior to expansion, and not less than 2290 mm (90 inches) in width. The cab protector shall cover the cab roof, fenders, and vehicle hood. Sufficient space shall be provided above the cab roof to allow for vehicle cleaning and maintenance.

* 3.5.2 Body mounting. A sub frame shall be provided. The turntable pedestal, outriggers and subframe shall be one integral welded or bolted (or a combination of the two) assembly. The subframe shall conform to the following:

When two outriggers are required or torsion bar stabilizers are required:

(a) The cabinet-loadspace floor mounting shall conform to 3.5.2.2 through 3.5.2.2.3 and the subframe length shall be at least sufficient to evenly distribute the pedestal and outrigger loads on the chassis frame rails or:

(b) The cabinet-loadspace floor mounting shall conform to 3.5.2.3 and the sub frame shall be the full length of the chassis frame to the rear of the cab.

When four outriggers are required:

(a) The cabinet-loadspace floor mounting shall conform to 3.5.2.2 through 3.5.2.2.3 or to 3.5.2.3, both with a subframe the full length of the chassis frame to the rear of the cab.

3.5.2.1 No welding of chassis frame. No part of the body (cabinets, floor, turntable pedestal, outriggers or subframe) shall be welded to the chassis frame.

3.5.2.2 Flexible mounting for cabinet-floor assembly. The cabinets and the loadspace floor shall be an integral assembly, secured independently of the turntable pedestal and the outriggers to the chassis frame or to the subframe with U-bolts, twin studs or brackets and shall include a wood breaker strip. All but the rearmost cabinet-floor assembly mountings shall be spring loaded to provide for movement of the cabinet floor assembly independent of the subframe, chassis frame, turntable pedestal, and outriggers.

3.5.2.2.1 U-bolts or twin studs. When U-bolts or twin studs are used for mounting the cabinet-floor assembly, there shall, be not less than four U-bolts or twin studs per aide having a 14 mm (0.563 inch) minimum body diameter with a 16 mm (0.625 inch) minimum thread diameter. Tie-plates shall be at least 13 mm (0.5 inch) thick. The vehicle chassis frame and the subframe shall be braced, using wood blocks, at each mounting point unless the mounting point is located at a full depth crossmember. Blocks shall incorporate a keeper strap or groove for the

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mounting bolt and shall be of a width to assure retention. Two tieback straps shall be provided, one bolted to each aide of the rear, to maintain cabinet-floor assembly alinement on the chassis frame rails or subframe.

3.5.2.2.2 Brackets. When brackets are used, they shall be bolted to the web of the subframe or to the chassis frame rails. The body mounting brackets shall provide means for drawing down the body on the subframe or chassis rails and provisions shall be made to prevent lateral shifting of the breaker strips. When additional holes are required to secure the mounting brackets to the chassis frame rails, they shall be located within the area of the rail which is designated as being safe for drilling in accordance with the chassis manufacturer's body builders layouts. Attachments shall not interfere with nor obstruct existing chassis components.

3.5.2.2.3 Breaker strips. A hardwood or dense southern yellow pine breaker strip not less than 29 mm (1-1/16 inches) in finished thickness shall be installed between the longitudinal sills of the cabinet-floor assembly and the subframe. Breaker strips shall have a taper of 25 mm (1 inch) in 460 mm (18 inches) at the forward end.

* 3.5.2.3 Rigid mounting for cabinet-floor assembly. The cabinets shall be bolted and the loadspace floor shall be bolted or welded to the full length subframe. The full length subframe and the chassis frame shall be tied together in such a manner as to act as a single member resisting bending and distortion of the cabinets. The section modulus of the full length subframe and the chassis frame combined shall be not less than 1150 cubic centimeters (70 inches cubed) per frame rail. A wood breaker strip conforming to 3.5.2.2.3 shall be provided when the cabinets and the loadspace floor are an integral assembly. When the cabinets are mounted independently of the loadspace floor, wood or metal breaker strips shall be provided when needed to provide relief at rivet heads, bolt heads, or other subframe irregularities.

3.5.3 Cabinet. Cabinet backs, partitions and shelves shall be constructed of not less than 20 gage (0.919 mm) (0.0359 inch) high tensile steel or 18 gage (1.214 mm) (0.0478 inch) carbon steel, reinforced. Cabinet doors shall be double panel construction of not less than 22 gage (0.759 mm) (0.0299 inch) steel. Doors shall be weatherproof and constructed to maintain fit and alinement in service. Doors shall have continuous rod or hem type heavy duty hinges, with stainless steel hinge pins and stainless steel, nylon or bronze insert sockets. Latches with recessed handles shall be furnished for all doors. Cylinder type locks, keyed alike, shall be installed in each door. Moving parts of latches and locks shall be constructed of, or plated with corrosion-resistant material. Cabinet doors over 1168 mm (46 inches) in length or height shall have pillar bolts or two latches. Dropdown doors shall have corrosion-resistant stay chains and shall be reinforced for use as shelves capable of supporting a live load of not less than 120 kg/m²

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(25 pounds on any square foot) of area. The cabinets and shelves shall be capable of supporting a live load of 490 kg/m² (100 pounds on any square foot) of area.

3.5.3.1 Cabinet arrangement. Vertical cabinets shall have minimum lengths of 460 (18), 610 (24) or 760 mm (30 inches), and horizontal cabinets shall have a minimum length of 1250 mm (49 inches) and shall conform to the following:

(a) Full height front vertical cabinets shall be a minimum of 380 mm (15 inches) in depth, and shall be equipped with bolted on, or screwed on, rope hooks
(b) Full height rear vertical cabinets shall be a minimum of 380 mm (15 inches) in depth
(c) Intermediate vertical cabinets shall be a minimum of 380 mm (15 inches) in depth
(d) Horizontal cabinets shall be located over the wheelwell housings and shall be a minimum of 380 mm (15 inches) in depth, 510 mm (20 inches) in height and 1250 mm (49 inches) in length
(e) Horizontal cabinets shall be equipped with small parts bins with adjustable spacers, shelves and partitions, and a rubber goods compartment, which shall be insulated and vented. Vertical cabinets shall be equipped with not less than two adjustable tray type shelves. The water cask, when required (see 3.5.5.1), shall be mounted in a curbside vertical compartment. An insulated hot stick compartment, for accommodating hot sticks at least 3050 mm (10 feet) in length, shall be furnished, except on class A.

3.5.3.2 Front bulkhead. The front bulkhead shall be capable of withstanding a horizontal static load equal to half the payload capacity of the vehicle without permanent distortion.

3.5.4 Lights. In addition to the lighting specified in 3.4.2.4, the following lighting and wiring shall be furnished:

(a) Lights with toggle switches and guards in each cabinet compartment
(b) Lights on universal mounts and of sufficient candle power to illuminate body loadspace and all equipment contained therein and all controls
(c) Two 115 mm (4.5 inch) minimum diameter sealed beam floodlights with universal mounts shall be installed, one on each side on the top of the cab protector, with controls on the outside and operable by an operator from inside the cab through an open window for illuminating the work area

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(d) Wiring for the body shall be in accordance with the applicable portion of the General Section, SAE J1292 and all wiring exposed to possible physical damage, such as wiring on a compartment floor, or wiring where tools or materials may cause the wire to be frayed or shorted, shall be protected by the use of heavy wall thermoplastic conduit or steel conduit having waterproof connections. Grommets shall be used for all wiring going through any hole or body framework

(e) An amber, rotating, vehicular warning light, in accordance with MS 51317-4, shall be provided. The rotating light shall be mounted with reinforcement at the point of attachment on or near the left front corner of the cab guard for 360-degree visibility. The rotating light shall be operated by means of a separate switch, with red indicator light located on the instrument panel and marked to indicate the function of the switch

(f) A demountable projecting load light shall be furnished on the end of the boom, with service line and socket, when the end of the boom extends beyond the rear of the vehicle. A receptacle on the rear of the body for the light shall be provided. The light shall be in accordance with Federal Motor Carrier Safety Regulation 393.18.

3.5 Attached accessories.

* 3.5.5.1 Vise. A vise shall be provided. The vise shall have nominal 3-inch hardened replaceable general material jaws and hardened replaceable pipe jaws. The vise shall be of the swivel type and shall be bolted to the right rear corner of the platform extension.

3.5.5.2 Water cask. When specified (see 6.2), a standard commercial 11.4 L (3-gallon) insulated potable water cask shall be furnished. The inside liner shall be stainless steel or non-toxic poly material. The outside shell shall be galvanized steel or equivalent. The throat area shall be of sufficient size to allow cleaning the interior of the water cask. A nylon or brass chrome-plated spigot, flush mounted to damage, shall be furnished. Lugs and bracket(s) shall be provided to mount the cask (see 3.5.3.1). Means shall be provided to retain the water cask and to minimize its vibration during road travel. Means shall be provided to drain excess water to the ground. A cup dispenser for dispensing 177 milliliter (ml) (6 ounce) cold vending cups shall be furnished. The dispenser shall be installed in the cabinet in a location accessible from the water cask.

3.5.5.3 Portable gasoline engine driven generator. When specified (see 6.2), a portable gasoline engine driven generator shall be furnished. The generator shall have a power rating of 120/240 volts, 60 hertz, single phase, 3-wire and shall provide not less than 3500 W. The generator and its circuitry shall conform to NFPA No. 70. Tiedown devices to secure the generator to the left front of the loadspace floor shall be furnished.

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3.5.5.3.1 Receptacles. The generator shall be equipped with two receptacles: one U-ground, 120-volt, 30-ampere and one three-wire, twist-lock type, 240-volt, 15-ampere.

3.5.5.3.2 Engine for generator. The engine for the generator shall be an air cooled, four cycle, magneto ignition gasoline engine having not less than 5.2 kW (7 horsepower). The engine fuel tank shall have a capacity of not less than 3.78 L (one gallon). The fuel tank shall have a drain and an inlet valve to the engine.

* 3.6 Aerial tower. Aerial tower structural components shall have a calculated design stress not greater than that allowed by ANSI A92.2. The aerial device shall be designed and constructed in its entirety in accordance with ANSI A92.2.

3.6.1 Turntable. The turntable assembly shall be rotated by a hydraulically operated motor. The turntable shall be so constructed as to eliminate all side thrusts on the motor shaft.

* 3.6.2 Fiberglass reinforced plastic. Fiberglass booms, basket(s) and components shall consist of extruded, molded, or spiral wound fiberglass. Each component shall conform to the performance requirements for electrical insulation specified herein. All components shall be made from a combination of multiple layers of sterile electrical grade fiberglass roving cloth, mat or strands with high impact strength resins with weathering and fire-retardant characteristics. The insulated booms shall have a hard, smooth surface inside and outside to prevent contamination and to preserve dielectric characteristics. When gelcoat is applied, the thickness of the gelcoat shall be not less than 0.16 mm (0.006 inch) and not more than 0.76 mm (0.030 inch). The parts shall be free from uncured areas, gaps, cracks, holes, blisters, fuzzy surfaces, resin pockets, areas lacking resin, laps, wrinkles, delamination, air or gas pockets, patches, porosity and other similar defects. The entire surface shall be smooth, fully cured and free of tackiness. Percentage of glass by weight shall be approximately uniform. The plastic construction insulation shall meet or exceed the specified performance requirements in 3.6.3.

* 3.6.3 Dielectric properties. The complete aerial platform (aerial device) shall conform to and shall pass the design, quality assurance and qualification tests required for a category A or category B aerial device rated at 69 kV in accordance with ANSl A92.2, without exceeding the maximum allowable boom currents specified in ANSI A92.2 for a 69 kV rating. Any mechanism within or along the insulated boom(s) shall be made of insulating materials, or so insulated as to demonstrate the same dielectric properties as the boom(s).

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* 3.6.3.1 Insulating insert, lower boom. An insulating insert shall be provided for the lower boom of size 30 through 85 vehicles. The insulating insert shall withstand a minimum of 50,000 volts ac, 60 Hz, with a leakage of not more than 1.0 milliampere. There shall be no evidence of dielectric puncture.

* 3.6.3.2 Dielectric test electrode(s). Test electrode(s) shall be installed on the insulated boom section(s) in accordance with ANSI A92.2. Test electrode(s) shall consist of two collector bands, to match the inner and outer contours of the insulated boom section(s). The test electrode(s) and collector bands shall be arranged to enable monitoring dielectric leakage of the upper fiberglass boom, hydraulic hoses, hydraulic fluid and leveling devices. Collector bands shall be of a conductive material secured to the insulated boom section(s). The electrode(s) shall incorporate a terminal for test circuit connection. A dual-range meter of 2.0 milliampere and 200 microamperes shall be furnished in a leather or heavy duty vinyl carrying case and stowed in a protected location for use by the operating personnel. An instruction plate shall be provided and shall include the maximum dielectric leakage permitted and shall describe the dielectric teat procedures. The basket liner(s) and booms shall be tested in accordance with ANSI A92.2.

3.6.3.3 Basket dielectric properties. Basket(s) shall withstand a minimum of 30,000 volts ac, 60 Hz, without evidence of dielectric puncture (see 6.6). A decal stating "NO INSULATION VALUE WITHOUT LINER" shall be applied on each basket. The liner(s) shall withstand a minimum of 50,000 volts ac, 60 Hz without evidence of dielectric puncture.

* 3.6.4 Boom assembly. The boom assembly shall be an articulated folding arm type and shall be hydraulically operated. The boom assembly shall be attached through the rotating turntable to a pedestal using self-alining bearings and shaft or shall be machined to assure alignment of the shaft and bearing surfaces. Under dynamic conditions, the boom assembly shall support the specified load placed in the aerial basket(s) in any extended position without permanent deformation of any component in or along the boom(s) or its supporting structure.

* 3.6.4.1 Boom construction. The booms shall be designed, engineered and constructed of materials that display the minimum weight conducive to the configuration and maximum performance requirements specified herein. Hydraulic lines and other associated parts which are fastened to, or inside, the booms shall be positioned for easy inspection, ready maintenance or possible replacement and shall have maximum protection against damage afforded by the configuration of the booms. The booms shall be attached to the aerial platform and to the turntable by replaceable shafts. Rotating bearing surfaces of the lower boom at the turntable and at the location of the lower and upper booms which are subjected to continuous wear shall be provided with replaceable bearings or bushings. Lubrication fittings or permanently lubricated

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components shall be provided on all rotating parts. Bearing surfaces shall be adequate to resist torsional moment when inherent within the design. When completely closed booms are furnished, means shall be provided within the booms to absorb moisture. Booms shall be provided with inspection openings at critical areas. Booms wrapped for insulation purposes shall not be acceptable.

3.6.4.2 Boom travel of lower arm. The boom travel of the lower arm shall be a minimum of 80 degrees vertical to the body platform.

3.6.4.3 Boom travel of upper arm. The upper arm boom angular travel for type I vehicles shall be not less than 125 degrees. Upper boom range of angular travel between booms for type II vehicles shall be not less than 200 degrees. Cable and sheaves, when furnished, shall be compatible with each other and shall have sufficient structural strength to meet the performance requirements specified herein.

3.6.4.4 Boom operation. The rotating turntable and boom assembly shall be hydraulically actuated by separate mechanisms and shall operate independently, or in any combination of raising, lowering and rotating at the same time. When in combination, hand metering of the controls may be necessary. The boom(s) operating mechanism shall be capable of limiting the lifting capabilities from rest position within safe limits of design stress to prevent overloading. For testing purposes, means shall be provided in the hydraulic system to perform the overload tests. Upon completion of all testing, the hydraulic system pressures shall be adjusted to the manufacturer's recommended setting for the model vehicle furnished. The vehicle shall be inherently designed or provided with a device to prevent the booms from reaching a position that would cause the vehicle to become unstable.

3.6.4.5 Boom storage. For normal storage, travel and shipment of the vehicle, the booms shall fold to essentially horizontal positions in the vertical plane parallel to the longitudinal centerline of the vehicle. Means shall be provided to hold down the boom assembly and platform to prevent unactuated raising of the booms when in transit. Saddles and a tiedown device shall be provided to support and prevent vertical and side movement of the boom and basket assembly in transit. Boom saddles shall be sufficiently padded to prevent damage to the booms. Supports shall not interfere with entrance and exit of personnel to and from the body loadspace. Maximum stowed height shall conform to 3.2.9.

3.6.4.6 Boom cover(s). Boom cover(s) and means to retain cover(s) on insulated sections of the booms while in stowage or storage-in-transit position shall be provided. The cover(s) shall completely enclose the insulated arm(s). Provisions shall be made to stow the cover(s) in a compartment of the body.

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* 3.6.4.7 Hydraulic jib and winch. A non-conductive hydraulic tilting jib and a hydraulic winch shall be furnished. The jib shall hydraulically tilt from 0 degrees or a negative angle to the axis of the boom to 45 degrees above. The jib shall manually extend from not more than 1070 mm (42 inches) to not less than 1520 mm (60 inches), measured from the hinge point of the jib to the load line. Quick disconnect tool outlets shall be mounted at the boom tip. The jib shall be of fiberglass. The winch shall conform to SAE J706, except an automatic safety brake is not required if the winch has a self-locking worm gear drive, capable of continuously holding and safely lowering a load equal to not less than 1.2 times rated winch capacity. The winch shall be wound with not less than 13 mm (0.5 inch) diameter insulated rope, consisting of braided polyester fibers with a braided nylon cover. The rope shall be of a length to reach the ground at maximum boom elevation height. The rope shall be provided with a spliced eye and quick hook. The capacity of the entire system shall be not less than 680 kg (1,500 pounds). Complete controls shall be provided within easy reach of the operator in the basket.

* 3.6.4.8 Lifting attachment for conductors. When specified (see 6.2), a lifting attachment to raise conductors off insulators shall be furnished. The attachment shall be capable of lifting at least three conductors simultaneously by elevating the upper boom. The attachment shall be secured to a hydraulic jib furnished in accordance with 3.6.4.7. The attachment shall be provided with a mast support and a quadrant. The mast shall be of fiberglass and shall meet the requirements of 3.6.3. An insulated auxiliary cross arm, a pivoting cross arm clamp, a brace, and three wire holders shall be provided.

3.6.5 Aerial basket(s). The total load capacity of the aerial basket(s) shall be not less than 227 kg (500 pounds) for two-man capacity and 136 kg (300 pounds) for one-man capacity. Strength of material of the basket(s) shall be as specified in 3.6.

* 3.6.5.1 Fiberglass basket(s). A one-man basket shall be furnished for sizes 30 and 35, type I and type II. Unless otherwise specified (see 3.6.5.2), a single, two-man fiberglass basket shall be furnished on type I vehicles, size 35 and larger. Two individual one-man baskets shall be furnished for all type II vehicles. The basket(s) shall be enclosed on four sides and the bottom. The basket(s) shall have a molded-in or bonded-on step, protruding to the outside, located to aid entrance and exit of the operator. The basket minimum interior dimensions shall be as specified in table XIV. Minimum interior dimensions shall be measured at the bottom of the basket(s) just above the bottom radii. Non-rectangular baskets may be furnished provided the total usable floor area is equivalent to that of a rectangular basket with the minimum dimensions specified in table XIV.

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TABLE XIV. Fiberglass basket(s) dimensions.

Dimensions	Two-man basket		One-man basket	
	mm	inches	mm	inches
Depth to floor, inside	1020	(40)	1020	(40)
Width	1156	(45.5)	546	(21.5)
Front to back	546	(21.5)	546	(21.5)

3.6.5.1.1 Basket liner(s). Except as otherwise specified (see 3.6.5.2), a nonconductive basket liner shall be provided for each basket furnished. The liner shall cover the basket interior. The liner shall be easily removable from the basket(s) for cleaning and for testing the dielectric capacity. The floor of the liner shall be provided with a removable, nonconductive scuff mat.

3.6.5.1.2 Basket cover(s). Waterproof basket cover(s) and means to retain cover(s) on the basket(s) while in stowage, or stowage-in-transit position, shall be provided on all basket(s) and also may enclose the controls; otherwise cover(s) shall be provided to enclose all control mechanisms completely. Provisions shall be made to store the cover(s) in a compartment of the body.

3.6.5.2 Guardrail type basket. When specified (see 6.2), a non-insulated guardrail type basket shall be furnished. The basket shall have two guardrails and a kickrail provided around the perimeter of the basket. The kickrail shall be not less than 13 mm (5 inches) in height. The lower and upper guardrails shall be not less than 510 mm (20 inches) and 1070 mm (42 inches) respectively from the floor of the basket. The flooring shall be of the open grating type. Interior dimensions shall be not less than 1070 mm (42 inches) in depth, 1020 mm (40 inches) in width and 760 mm (30 inches) front to back. A gate with a positive safety latch to hold the gateway closed shall be provided. Protruding edges shall be padded with a resilient replaceable bumper. The guardrails shall be constructed of aluminum.

3.6.5.3 Flooring. Basket(s) shall have a reinforced floor. No point in the flooring shall deflect more than 6.4 mm (0.25 inch) nor be permanently deformed when subjected to a vertical load of 166 kg (300 pounds) covering an area of 160 cm^2 (25 square inches).

3.6.5.4 Safety belt(s). Lineman type safety belt(s) with lifeline(s) and positive lock type, quick release feature(s) shall be furnished for each personnel space in each basket. Each belt shall be capable of withstanding a static load of 1815 kg (4,000 pounds) without damage or permanent deformation and shall conform to the Occupational Safety and Health Standards (OSHA), 29 CFR. The safety belt(s) shall be designed and positioned to allow freedom of movement of the workman in the basket(s). Means shall be provided to attach fasteners to the end of the boom or basket(s) bracket and to prevent fastener attachment being pulled through the fiberglass wall in the event of leveling system failure. All fasteners and attachments shall be corrosion-resistant.

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3.6.5.5 Tool trays. One fiberglass or polyethylene tool tray for each personnel space in the basket shall be provided to fit over the edge of the basket(s). Provisions shall be made to secure and stow the trays in a compartment of the body.

* 3.6.5.6 Basket(s) engine starter and cutoff. An engine starter and cutoff for the truck engine, remotely operated from the basket(s), shall be furnished. The engine starter and cutoff shall not reduce the insulating characteristics specified in 3.6.3.

3.6.5.7 Rotating basket(s). When specified (see 6.2), rotating basket(s) having not less than 45 degrees of rotation in either direction about their vertical axis, transverse to the upper boom, shall be furnished. Total rotation shall be not less than 90 degrees. Gear rotation shall be self-centering in final drive and controlled from the basket(s). When two baskets are furnished, the basket with the controls may have restricted rotation in one direction to prevent the controls from hitting the boom.

3.6.5.8 Basket leveling system. The basket leveling system shall be either cable, roller chain, or hydraulic. The leveling system shall be designed with minimum stretch due to loading of the basket(s). The system shall be capable of keeping the basket(s) level with the rated load in the basket(s) positioned on the floor of the basket(s). An adjustment shall be provided to maintain horizontal position of the basket(s). The system shall be insulated as specified (see 3.6.3). The system shall have a minimum insulation of 2740 mm (9 feet). Two independent leveling systems shall be furnished. The secondary system may be gravity, where the bottom of the basket(s) shall stay down in case of failure. Baskets suspended such that failure of the primary leveling system will result in gravity tipping the loaded basket shall be provided with a secondary non-gravity leveling system. The leveling system shall be automatically operated in all positions the booms are capable of attaining. The basket shall be capable of being released and easily tilted over, under control by one man, near ground level, manually or hydraulically, to remove a disabled operator or to dump out water.

3.6.6 Controls. A complete set of hydraulic hand operated controls for maneuvering the basket(s) shall be provided both at the basket(s) and at the base of the lower boom. Except for sizes 100 and 120 and larger, a single lever control for all maneuvering operations shall be furnished at the basket control station. The placement of the controls shall be so as to prevent inadvertent actuation by the operator. Controls for maneuvering the basket(s) shall be located at the right or left side of the basket(s) in a position easily accessible to the operator facing the work area. Controls shall not be mounted on the front or rear of the basket(s). The ground controls at the base of the lower boom shall be easily accessible from the floor of the body, otherwise a platform shall be provided for the operator at the base of the boom. Ground controls shall be placed in such a manner as to minimize the possibility of use by unauthorized personnel.

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3.6.6.1 Operation. Smooth operation, with no lurching of the boom assembly, shall be possible. The controls shall be self-centering and shall revert to a neutral position when released by the operator. All boom and outrigger controls (see 3.6.7.1.1) shall be of the dead-man type operable with one hand but whereby two separate actions must be taken to actuate the control. The controls at the base of the lower boom shall be capable of overriding the controls on the aerial basket(s).

3.6.6.2 Instruments. Any switches, instruments and pressure gages on the lower control station shall be panel mounted, identified as to their function, and installed in a manner to facilitate removal and servicing. Gages and instruments shall be shockproof, waterproof, and weatherproof. Identification markings on corrosion-resistant plates shall be provided.

3.6.7 Stabilizing equipment.

3.6.7.1 Outriggers. Two hydraulically operated outriggers shall be provided if the actual maximum attainable floor to ground height of the basket is less than 9750 mm (32 feet) for size 30 and less than 12 m (36.5 feet) for size 35 (class A vehicle may be provided with torsion bar stabilization in lieu of outriggers, provided it meets all the requirements of 3.3.3). Four hydraulically operated outriggers shall be provided if the actual height is 12 m (36.5 feet) or greater or, for size 30, greater than 9750 mm (32 feet). Outriggers shall:

(a) enable leveling the vehicle on inclined surfaces and
(b) shall insure adequate stability with relation to the ground while operating the aerial platform under maximum loads.

Outrigger cylinder hydraulic fittings, lines, tubing, hoses, and pilot check valves shall be protected by metal guards, but shall be accessible for servicing. Outriggers shall be designed so that they may be extended over curbings and sidewalks.

* 3.6.7.1.1 Outrigger controls. An individual control for each outrigger shall be provided. Controls for the left outriggers shall be located at the rear left corner of the vehicle and controls for the right outriggers shall be located at the rear right corner of the vehicle. All outrigger controls shall be installed in a readily accessible location allowing operation from ground level. An automatic pilot check valve shall be provided for each outrigger to block the hydraulic fluid in the cylinder and to prevent movement of the outrigger in the event of a damaged hydraulic line. Interlocks shall be furnished to prevent the operation of the aerial boom assembly until the right and left side outriggers have properly descended. A self-actuating locking device shall prevent accidental raising of the outriggers while the aerial boom is elevated from stowage position. The interlocks shall not interfere with the operation of any hydraulic tool circuit. Each outrigger shall be provided with an automatic, positive, spring-actuated lock, or an automatic

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pilot check valve which will hold the outrigger in the retracted position without settling and which may be manually released.

3.6.7.1.2 Outrigger pads. Each outrigger shall be provided with a ground pad. Ground pads shall be of the hinged or fixed type. Each pad shall have a ground contact surface of not less than 929 cm^2 (144 square inches) for sizes 30 through 50, and not less than 2065 cm^2 (320 square inches) for all other sizes. In order to provide the 2065 cm^2 (320 square-inch) ground contact per pad, a portable pad for each outrigger may be furnished to be used in conjunction with manufacturer's standard pads. The portable pads shall be steel plates equipped with a handle. The vehicle body shall be equipped with secure stowage provisions for the portable pads. The stowage provisions shall be designed so as to prevent loss of the pads due to vehicle motion.

3.6.7.1.3 Outrigger warning light. A red warning light shall be installed in the cab in a conspicuous location to warn the driver when the outriggers are extended.

3.6.7.1.4 Circular level. A circular level, with 0 to 5 degrees reading, shall be furnished. The circular level shall be mounted in a protected position, visible from all outrigger controls. The circular level shall indicate the degree of body level as the outriggers are maneuvered. The circular level shall be corrosion-resistant and resistant to inclement weather. A corrosion-resistant warning plate shall be mounted adjacent to the level to indicate the maximum longitudinal incline and the maximum transverse slope in degrees on which the vehicle is capable of safely operating.

3.6.8 Hydraulic system. A hydraulic fluid, nonconductive system consisting of all components required to perform all hydraulic operations of the boom assembly and outriggers with the degree of control specified herein shall be provided. The hydraulic system shall also include the components required to power the front mounted winch, when required (see 3.4.22). Mechanisms and reservoirs shall be positioned or shielded to prevent hazardous fluid contact with heated surfaces or electrical apparatus. A two-speed manual or an automatic hydraulic throttle actuator shall be furnished which increases the speed of the truck engine (and auxiliary engine, when furnished) when the pump is being loaded. An accumulator may be used as a component of the hydraulic system to provide the maximum degree of temperature control under continuous service. All hydraulic components shall comply with ANSI A92.2.

3.6.8.1 Hydraulic cylinders. All hydraulic cylinders shall incorporate seals to prevent entrance of moisture and dirt into the cylinders and leaks of hydraulic fluid from the cylinders. Preformed packings shall be used in the cylinders where static seals are required. Manufacturer's standard seals shall be furnished where applied to reciprocating parts. All

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hydraulic cylinders for actuating the booms and outriggers shall be provided with pilot check or holding valves to prevent descent of the boom assembly and retraction of the outriggers in the event of loss of hydraulic pressure. Means shall be provided to install pressure gages for testing the hydraulic system pressure and relief settings. All surfaces of hydraulic ram piston rods shall be chrome plated with sufficient thickness to withstand a 30-hour salt spray test in accordance with ASTM B 117.

3.6.8.2 Hydraulic lines and fittings. Hydraulic lines shall be located for maximum ease of removal and installation, or visual inspection for failure, and maximum protection from damage during operation of the boom assembly. Flexible hydraulic hose or collector rings shall be used where flexing of hydraulic lines is encountered and where installation of components requires a flexible coupling. Standard straight thread, flared type fittings, fittings adapted to flared type, or tapered pipe thread with dry seal fittings shall be furnished. Hydraulic lines and fittings shall be air- and liquid-tight and shall be installed to prevent vibration and chafing. The system shall be designed to prevent hydraulic transients. Provisions shall be made for bleeding air from the hydraulic system and completely draining all hydraulic lines, when necessary. A manual type bleeder shall also be provided as on-vehicle equipment on closed-center, non-self bleeding systems. Hose assemblies and all seamless steel hydraulic line shall be adequate to provide the rate of flow, temperature and pressures required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be acceptable.

* 3.6.8.3 Hydraulic pump. A hydraulic pump with necessary pressure and flow capacities to operate all hydraulically powered components shall be provided. The pumping system shall include an adjustable pressure relief valve, unloading valve or pressure compensator with a range of adjustment which allows raising or lowering of the boom(s) and basket(s) with any working or test loads specified herein. A readily visible pressure gage shall be provided on the discharge side of the hydraulic pump.

* 3.6.8.4 Pump drive systems. The hydraulic pump shall be driven by a heavy duty power takeoff unit attached to the truck chassis transmission. For emergency use, except when a rechargeable electrical or auxiliary engine driven system is specified, an auxiliary hydraulic pump, driven by an electric motor powered from the vehicle chassis electrical system, shall be provided to rotate the turntable, retract the outriggers, lower the booms and place the booms in stowed position in the event of primary power source failure. The system shall be equipped with cut-in devices and controls at the basket(s) and at the lower control station. The system shall provide adequate pressure and flow capacities for operation when the power takeoff driven power source fails. When specified, in addition, to the power takeoff, the hydraulic pump shall be driven by a rechargeable electrical system (see 3.6.9), or by an auxiliary engine (see 3.6.10). Means of rapid conversion from one power source to the other, within the basket and from the

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ground, shall be provided. When an air compressor is specified (see 3.6.19.9), the hydraulic pump and the air compressor shall be driven by an auxiliary engine. Chain drives shall not be used in any system.

3.6.8.5 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided, with sufficient capacity to hold the required volume of hydraulic fluid, with not less than 10 percent for expansion space. The reservoir shall be baffled and suction and reservoir ports shall be located as necessary to ensure thorough mixing of hot return oil and as necessary to prevent foaming oil from entering the hydraulic system suction port. The reservoir shall be provided with a filler opening, a removable fill opening oil strainer and a vented filler cap, situated in an accessible location to facilitate filling the tank from a one-gallon container using a funnel. The vented filler cap shall have means to prevent fluid from splashing out of the tank during movement of the vehicle. The filler cap shall be attached to the tank by a corrosion-resistant chain or cable. Provision shall be made for inspection and cleaning of the interior and for draining the reservoir. The reservoir drain shall be located so that fluid will not drain on other components and shall allow recovery of the fluid in a container. The reservoir shall have a corrosion-resistant plate to indicate the type of hydraulic fluid furnished and the maximum fluid level in the reservoir with the booms folded for travel and the hydraulic lines and components full of fluid. A sight level or dipstick gage shall be furnished to observe the level of fluid. The tank shall be clean.

3.6.8.5.1 Hydraulic fluid. Unless otherwise specified, hydraulic fluid conforming to the manufacturer's standard or recommended requirements shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system.

* 3.6.8.6 Oil filter and strainers. A full-flow type, 10-micron discharge oil filter in the return line to the reservoir or in the pressure line immediately after the pump shall be provided. A 100-mesh line strainer shall be provided on the auction aide of the hydraulic system. The filter and the auction strainer shall be easily accessible for periodic cleaning, or for replacement of the filter cartridge. The filter and the strainer shall have sufficient capacity to allow the pump to deliver maximum rated flow.

3.6.8.7 Hydraulic fluid temperature. The hydraulic fluid temperature shall be maintained within safe limits to prevent seal damage or approach to the flash point of the hydraulic fluid during all operating conditions of the vehicle and use of the aerial tower hydraulic system for long periods of time. Excessive heating of the fluid (more than 13°C (55°F) above ambient in an ambient of 52ºC (125°F) shall be unacceptable under any operating condition. Provisions shall be made to prevent malfunction or unactuated operation of the hydraulic control system due to changes in fluid temperature.

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3.6.9 Rechargeable electrical system. When specified (see 6.2), a rechargeable electrical system shall be furnished to drive the hydraulic pump assembly, in addition to the normal power takeoff drive. The electric motor shall be mounted on the body platform in conjunction with the hydraulic pump. The motor shall have adequate output power for full compliance with all performance and lifting requirements and speeds specified herein. The mounting shall minimize transfer of vibration. The electrical system shall include:

(a) A minimum 12-volt dc motor
(b) Industrial battery(s) of not less than 450 ampere-hour capacity
(c) Provisions for charging battery(s) from the vehicle alternator (see 3.4.2.3)
* (d) A 220-volt, single phase battery charger with a dc output capable of recharging the battery(s) in (b) above in not more than 12 hours
(e) Hour meter
(f) Battery charge indicator
(g) Pressure switches and control circuits
(h) Battery isolator connected to the chassis charging system which prevents a drain of the chassis battery during boom operations
* (i) Connection of all vehicle lights to the rechargeable electrical system to enable operation of the lights when this system is being used.

3.6.10 Auxiliary engine. When an air compressor is furnished (see 3.6.10.9), an auxiliary engine to drive the air compressor and the hydraulic pump shall be furnished, in addition to the normal power takeoff drive. When specified (see 6.2), an auxiliary engine shall be provided to drive the hydraulic pump assembly only, in addition to the normal power takeoff drive. The auxiliary engine shall operate on the same type of fuel as the truck chassis engine. Unless an auxiliary engine fuel tank is specified, the auxiliary engine fuel supply shall be from the chassis fuel tanks. The engine shall be mounted on the body platform in conjunction with the hydraulic pump and, when applicable, the air compressor. Mounting shall minimize the transfer of vibration. The engine shall be capable of being started and stopped from the aerial basket. The auxiliary engine shall have sufficient output power to drive all operating components other than the vehicle. When a gasoline engine is furnished, it shall be capable of warranted operation on unleaded fuel.

3.6.10.1 Engine components. The engine shall be a two or four-cycle diesel engine on truck chassis with a diesel engine or a four-cycle gasoline engine on truck chassis with a gasoline engine. The engine shall be equipped with at least the following components:

(a) Distributor or magneto type ignition (gasoline engine only)
(b) Starter, 12-volt
(c) Resistor type spark plugs (gasoline engine only)

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(d) Air cleaner
* (e) Automatic choke (gasoline engine only)
(f) An engine hour meter, having a totalizing mechanism of not less than 9,999 hours
(g) When power plant heaters are specified (see 3.4.1.6), a 300-watt immersion type heater with 77ºC (170ºC) to 90ºC (195°F) thermostat shall be installed in the oil pan through any convenient opening.

3.6.10.2 Engine control panel. An engine throttle, a key-operated starting (ignition) switch, and a palm button starter shall be installed in a weather- and waterproof cabinet on a control panel, and each shall be identified as to its function. The cabinet shall be provided with a side- hinged cover with a spring type latch and provisions for locking with a 1-inch padlock.

3.6.10.3 Fuel pump. A fuel pump shall be furnished on the engine, lifting fuel against a head of 1219 mm (4 feet), the pump shall have sufficient capacity to supply the engine at its maximum rated load. An in-line fuel filter shall be provided.

3.6.10.4 Governor. A variable speed governor shall be furnished. The governor shall automatically control the engine speed at the selected rated speed at all fluid pressures and flows required. A manually-operated hand throttle shall control the engine at all speeds from the manufacturer's recommended minimum governed speed up to the maximum operating speed.

3.6.10.5 Auxiliary engine fuel supply. Unless an auxiliary engine fuel tank is specified (see 3.6.10.6), the auxiliary engine shall utilize the vehicle’s main engine fuel tank(s). A shut-off valve shall be provided in the fuel line in an accessible location. Fuel line(s) shall be in accordance with Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.6.10.6 Auxiliary engine fuel tank. When specified (see 6.2), an auxiliary engine fuel tank shall be furnished. The fuel tank shall have a capacity to provide fuel for at least 8 hours of full-load operation. The fuel tank cap shall be attached to the tank by means of a corrosion resistant chain or cable. A sump shall be provided in the fuel tank suitable for collection and accumulation of water. A drain shall be located at the bottom of the sump for complete drainage of the fuel tank. The drain shall be located to prevent spillage of fuel on any area of the vehicle. The tank filler shall be provided with a protective screen and vent. A shut-off valve shall be provided in the fuel line adjacent to the fuel tank. A corrosion resistant plate shall be furnished and marked denoting the type of fuel to be used, and shall be located in a conspicuous location near the filler cap. Fuel line(s) shall be in accordance with Federal Motor Carrier Safety Regulations 393.65 and 393.67.

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3.6.10.7 Auxiliary engine exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. Means shall be provided to vent exhaust fumes away from the cab and body. The exhaust system or hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the auxiliary or main chassis engine exhaust systems.

3.6.10.7.1 Auxiliary engine spark arrester. A spark arrester shall be furnished, having an 80 percent arresting efficiency when rated in accordance with SAE J350.

3.6.10.8 Auxiliary engine electrical system. The auxiliary engine electrical system shall utilize the truck chassis battery and shall be in accordance with Federal Motor Carrier Safety Regulations 393.30, 393.31 and 393.33.

3.6.10.9 Air compressor. When specified (see 6.2), an air compressor shall be furnished. The compressor shall be the two-stage type with constant speed control, shall have an actual displacement of 0.85 cubic meter per minute (30 cubic feet per minute), and shall provide a pressure of not less than 1.0 MPa (150 pounds per square inch (psi)). The air compressor shall be complete with receiver, moisture separator, automatic oilers, shock- and waterproof pressure gage, safety valve, and all necessary equipment for satisfactory operation of the compressor. The air compressor shall be powered by the auxiliary engine. The compressor shall be so installed as to minimize the transfer of vibration from the auxiliary engine. The receiver tank shall have a capacity of not less than 113 L (30 gallons). An air pressure regulator having a setting range of at least 344 kN/m² to 1030 kN/m² (50 psi to 150 psi) shall be furnished. An air line, extending from the compressor to an outlet in the cabinet at the curbside of the vehicle, shall be provided. An air line extending from the compressor to an outlet(s) at the basket(s) shall be provided and shall be of the nonconductive type. An air line shut-off valve shall be provided in the cabinet and at the basket outlets.

3.6.10.9.1 Pneumatic outlet. When an air compressor is specified (see 3.6.10.9), an air outlet shall be furnished in the basket(s). One 2440 mm (8-foot) air hose, for use in the basket(s), shall be furnished. The hose shall be rubber, nonconductive, with quick disconnect fittings on both ends, compatible with the outlet and standard pneumatic tools. Dust caps shall be furnished on both ends of the air hose. All critical pneumatic components shall comply with ANSI A92.2.

3.6.11 Hydraulic tools.

* 3.6.11.1 Ground hydraulic tool outlets. Unless otherwise specified (see 6.2), a hydraulic tool outlet shall be provided in the curbside cabinet or in a protected but accessible position on the rear, under the platform or on the curbside of the turntable pedestal. The tool outlet shall use

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hydraulic fluid pressure from the hydraulic system of the aerial platform. Discharge port(s) and return port(s) shall be provided with a corrosion-resistant cap, secured with a corrosion-resistant chain or cable, to protect the threads and the valve from contamination. A tool circuit with quick disconnect, dripless couplers shall be furnished for attaching the pressure and return hose lines for the motor driven hydraulic tools. The oil flow for the tools shall be regulated by control valve(s) to meet the flow requirements of the hydraulic tools. The high system pressure (working pressure) shall be not less than 1,600 psi and the back pressure in the return line shall not exceed 1030 kN/m² (150 psi).

3.6.11.1.1 Basket hydraulic tool outlet and hose. Unless otherwise specified (see 6.2), a hydraulic tool outlet shall be furnished in the basket(s). An 2440 mm (8-foot) dual hydraulic hose for use in the basket(s) shall be furnished. The hose shall be of the flexible, nonconductive type with quick disconnect dripless fittings on each end. Hose fittings shall be compatible with the basket outlet and the tools specified in 3.6.11.3 through 3.6.11.9. The hose ends shall be provided with dust caps.

* 3.6.11.2 Ground tool hose. Except when a ground hydraulic tool outlet is not required (see 3.6.11.1), or when a hydraulic hose reel is specified (see 3.6.11.11), two 15 m (50-foot) lengths of hydraulic wire braid hoses with quick disconnect, dripless couplers on both ends, for use at ground level, shall be furnished. Warning plate markings (see 3.1.1.2.1) shall warn that the wire braid hydraulic hoses shall not be used on or near electrical lines. The hoses shall be furnished with dust caps. Hose fittings shall be compatible with the tool outlet specified in 3.6.11.1 and the tools specified in 3.6.11.3 through 3.6.11.9.

* 3.6.11.3 Hydraulic capstan hoist. When specified (see 6.2), a hydraulic capstan hoist shall be furnished. The drum shall be one-direction drive and shall have a backstop on the output shaft to prevent reverse travel. (Lowering will be accomplished by letting the rope slip on the drum.) The capstan hoist shall be complete with a link chain and screw clamp attachment device for mounting to surfaces other than on a round pole. The hoist shall have a rated single line pull capacity of not less than 567 kg (1,250 pounds) on the bare drum. The hoist shall have a remote control, foot-operated, open-center valve.

* 3.6.11.4 Hydraulic impact wrench and drill. When specified (see 6.2), a hydraulic impact wrench and drill shall be furnished. This combination tool shall be reversible and capable of installing or removing lag screws and other hardware. The tool shall have a capacity for 1.063 inch drills, for 16 mm (0.625 inch) bolts, and shall include a 13 mm (0.5 inch) square drive socket. The tool shall be capable of developing not less than 200 foot- pounds of torque with at least 1,350 impacts per minute. A quick change drill chuck shall be furnished.

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3.6.11.5 Hydraulic chain saw. When specified (see 6.2), a hydraulic chain saw shall be furnished. The saw shall have a chipper type chain and shall be suitable for cutting trees up to 380 mm (15 inches) in diameter.

3.6.11.6 Hydraulic chain saw, pole mounted. When specified (see 6.2), a hydraulic chain saw, pole mounted, shall be furnished. The chain saw shall be light weight (approximately 4.5 kg (10 pounds)) and shall have a cutting capacity of 230 mm (9 inches). The pole shall be approximately 1520 mm (5 feet) long and shall have a fiberglass type body to provide good insulation.

3.6.11.7 Hydraulic circular saw, pole mounted. When specified (see 6.2), a hydraulic circular saw, pole mounted, shall be furnished. The saw shall have a capacity of not less than 81 mm (3.25 inches) and shall operate at approximately 2710 meters per minute (8,900 feet per minute). The pole shall be approximately 1520 mm (5 feet) long and shall have a fiberglass type body to provide good insulation. The saw shall include at least a 40-tooth blade.

3.6.11.8 Hydraulic pruner, pole mounted. When specified (see 6.2), a hydraulic pruner, pole mounted, shall be furnished. The pruner shall have a capacity of 50 mm (2 inches) in diameter. It shall be mounted on a fiberglass type pole to provide good insulation and shall be approximately 2130 mm (7 feet) long.

3.6.11.9 Hydraulic tool fittings. Quick disconnect, dripless fittings shall be provided on each hydraulic tool. The fittings shall be compatible with the hoses specified in 3.6.11.1.1, 3.6.11.2 and 3.6.11.11.

3.6.11.10 Hydraulic tool stowage. Hydraulic tools, when furnished, shall be securely stowed on brackets inside the curbside cabinets. The brackets shall be provided with retaining devices to secure the tools.

* 3.8.11.11 Hydraulic hose reel. When specified (see 6.2), a heavy duty retractable hydraulic hose reel and hose shall be furnished. The hose reel shall be furnished with a mounting plate secured in either a vehicle cabinet or in an accessible location in the vehicle loadspace. The hose reel shall be furnished with a weatherproof cover for protection when not in use. The hose reel shall be capable of withstanding not less than 110 percent of the hydraulic system working pressure. The hose is intended for use in the aerial basket and thus the hose and hose reel shall be capable of meeting the dielectric properties specified in 3.6.3. The hose reel shall be capable of retraction. The hose reel shall be marked, stating that the hose shall be walked back to the reel to prevent damage. Hose fittings shall be of the quick disconnect, dripless type and shall be compatible with the tool outlets specified in 3.6.11.1 and 3.6.11.1.1 and with the tools as specified in 3.6.11.3 through 3.6.11.9. The following accessories shall be furnished:

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(a) Flow control valve(s)
(b) Line check valve(s)
(c) Hose ball stop
(d) Four-way roller hose guide
(e) Dual hose, not less than 15 m (50 feet) in length, with dust caps.

3.6.12 Safety. All equipment or exposed portions of the equipment which create extreme temperatures, or that can be damaged by inclement weather, and all rotating or reciprocating parts which are of such a nature, or so located, as to become a hazard to operating personnel, shall be insulated, fully enclosed, or properly guarded.

3.6.12.1 Highway warning kit. Three emergency reflective triangles conforming to Federal Motor Carrier Safety Regulation 393.95 (f)(2)(i) shall be furnished and stowed in the cabinets.

* 3.6.13 Stress analysis. A stress analysis for each type, class, size and model vehicle under contract shall be performed. The stress analysis shall include the weight of all major items permanently installed on or within the vehicle, and shall indicate the location of the center of gravity of each component relative to the vehicle axles (travel and critical working positions). In addition, the analysis shall include the value of stresses at the base of the boom(s); at the hinge area between the upper and lower boom sections; at the hinge area between the upper boom section and the basket(s); and the center of gravity of each section of boom. The analysis shall verify the factors of safety specified herein, and shall verify that the yield point of the material used is not exceeded with the specified load on the basket(s).

3.7 Dissimilar metal. All dissimilar metals throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

3.8 Lubrication system. A lubrication system shall be furnished with fittings for all bearings and other working parts which require relubrication for continued service. The vehicle shall be lubricated prior to delivery with applicable grade of standard lubricant for the ambient air temperature encountered at the delivery point.

3.9 Workmanship. The quality and workmanship of all component parts, including fabricated parts, machined parts and castings, shall be commensurate with sound manufacturing standards for similar equipment. The vehicles shall be free from all burrs, sharp edges, die, tool and metal handling marks, rust, scale, and cracks. All metal and plastic materials used shall be free from delamination defects and shall be prime material. Improper manufacturing or assembling practices, defective components or parts, and assemblies which have been modified to overcome deficiencies, shall not be furnished.

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* 3.9.1 Welding. Adequate preparation of metal, prior to welding of the body and aerial tower components, shall be provided. Surface of parts to be welded shall be free of rust, scale, paint, grease and other foreign matter. Welding rods (electrodes) shall be compatible with the metal being welded. All critical arc welds shall be magnaflux or dye penetrant inspected. The following defects shall not be acceptable:

(a) Slug or plug welding
(b) Undersize of fillet welds, overlaps, inadequate fusion and penetration, undercutting of weldments, and vertical down welding
(c) Unremoved slag and spatter
(d) Cracks
(e) Spot, tack, or intermittent welds for strength.

Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Intermittent welding shall be permissible when in accordance with 3.9.2 through 3.9.3.

3.9.2 Workmanship specimens. Workmanship specimens shall be prepared by the manufacturer, for the body and aerial tower components, to simulate all welding conditions. The specimens shall represent the minimum acceptable weld quality and cleaning procedure to be used in production. Upon submittal of these specimens by the contractor, the quality level and identification shall be subject to approval by the Government. The specimens shall be prepared using the welding procedures to be employed in production. Specimens prepared to represent multiple pass welds shall be made in such a manner as to leave at least two inches of each layer of weld metal exposed. Approved specimens shall be retained at the contractor's plant.

3.9.3 Welding procedures. The welding procedure shall be recorded prior to production of any weldment, but no procedure qualification other than workmanship specimens shall be required.

3.9.4 Factors. The factors listed below which are applicable shall be included in the recorded welding procedures:

(a) Joint sketch
(b) Joint type and preparation
(c) Base metal type
(d) Filler metal, size and type
(e) Position of welding
(f) Arc-voltage, current range and polarity
(g) Manual or machine
(h) Travel speed (machine only)
(i) Pre-heat and post-heat.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First vehicle preliminary inspections.

4.3.1 Welding procedure and workmanship specimen evaluation. Prior to the start of fabrication of the sub floor, body, turntable, pedestal, outriggers, and boom assemblies, an inspection shall be conducted by the Government to evaluate workmanship specimens and welding procedures.

4.3.1.1 Macro-etching. Workmanship specimens shall be cross-sectioned and macro-etched. Both parts of the specimens shall be made available to the Government representative for examination.

4.3.2 In-process inspection of body and components. The first completely assembled body and components (subframe, turntable, pedestal, outriggers, boom assemblies, and basket) shall receive an in-process inspection by the Government prior to prime painting. The inspection shall

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be conducted to evaluate workmanship and to determine conformance of structural members to contractual requirements.

4.3.2.1 Structural inspection. The vehicle shall be tested in accordance with 4.3.2.2 through 4.3.2.4 prior to painting. After the tests, dye penetrant or magnetic particle inspection shall be performed on the critical areas of the welds, again, prior to painting. A water soluble penetrant inspection shall be performed on all critical plastic areas. The water soluble penetrant or magnetic particle inspections shall provide means to detect crack indications under dynamic and static conditions.

* 4.3.2.2 Dynamic conditions. With two times the specified payload (see 3.6.5) on basket floor(s) or suspended from the basket(s), the upper and lower booms shall be extended through various positions, as deemed necessary by the Government inspector, and with the turntable being rotated 360 degrees, for at least 1/2 hour, the water soluble penetrant and magnetic particle inspection areas shall be inspected for any evidence of cracks. A visual inspection shall be performed on all critical fiberglass components. Any evidence of cracks or deformation or of damaged fiberglass material shall be cause for rejection of the vehicle.

4.3.2.3 Static conditions. With the lower boom in its most vertical position, and the upper boom horizontal, and with 2-2/3 times the specified load (see 3.6.5) placed in or suspended from the basket(s), water soluble penetrant and magnetic particle inspection areas shall be inspected for any evidence of cracks. The vehicle shall be secured against instability during the test with 2-2/3 times the load. A visual inspection shall be performed on all critical fiberglass components. Any evidence of cracks or deformation or of damaged fiberglass material shall be cause for rejection of the vehicle.

4.3.2.4 Critical areas. At least the following critical areas shall be inspected by dye penetrant or magnetic particles:

(a) Outriggers and attaching areas
(b) Turntable, pedestal, base, trunnion and yoke at attachment of lower boom and struts
(c) Lower boom at cylinder clevis brackets
(d) Lower boom sheaves, sheave supporting brackets, cable anchor and support brackets
(e) Lower boom at attachment to upper boom
(f) Upper boom sheaves, sheave supporting brackets, cable anchor and support brackets
(g) Structural steel of upper and lower boom
(h) Upper boom and yoke at attachment to basket(s) and side brackets.

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4.4 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.1 Weight and payload capacity. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axles. The imposed loading on front and rear axles shall be computed by the contractor and verified by the Government to ascertain that the vehicle is adequate for the intended loading. Distribution of weight will be computed by using the curb weight, the operator weight at 175 pounds, and a payload considered to be evenly distributed throughout the length of the vehicle body to provide the specified GVW.

* 4.4.2 Road test. The vehicle, for 4x2 and 6x4 models and for models built to all wheel drive while on the chassis manufacturer's production line shall be road tested by the contractor with payload. The road test shall be for not less than 16 km (10 miles) at speeds up to 84 km/h (52 mph). The vehicle, for all wheel drive models qualifying under 3.1.1.13 shall be road tested with and without payload. Payload shall be distributed first, so that the front axle is loaded to GAWR and the vehicle is loaded to GVW and second, with the rear axle loaded to GAWR and the vehicle loaded to GVW. The road test for each of the three conditions shall be for not less than 48 km (30 miles) at speeds up to 84 km/h (52 mph) over highways and gravel roads and for not less than 8 km (5 miles) at speeds up to 24 km/h (15 mph) over cross country terrain with ground and grade requiring all wheel drive. During the loaded and empty portions of the road test, the brakes shall be applied firmly, bringing the vehicle to a sudden stop not less than 5 times during each portion of the road test. During the road test, the front and rear suspension and the drivetrain shall be periodically inspected for interference and contact with other vehicle components. Abnormal contact of the drivetrain or suspension components with other components shall be cause for rejection. Front spring bumpers shall not make contact with frame stops except under the most extreme cross country conditions. Operational or mechanical failures of the vehicle components during the road test shall be cause for rejection. Failure includes permanent deformation as well as breakage.

4.4.3 Truck body treatment and painting. The certification regarding the body cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be made to Government representatives for examination and approval.

* 4.4.4 Structural verification. The contractor shall present, at the time of the first production vehicle inspection, test results from the in-process inspection (see 4.3.2 through 4.3.2.4) or shall

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repeat the in-process inspection. Upon Government verification that the tests and inspections were successfully conducted and witnessed by the Government, the tests of 4.3.2.1 and 4.3.2.2 need not be repeated.

4.4.5 Upsetting and range of movement test. With the truck resting on approximately level ground, without payload, and not fastened to any artificial base, the basket(s) shall be loaded with two times the specified payload (see 3.6.5). The basket(s) shall be extended to its full reach and exercised through the full range of traverse and elevations indicated in 3.3.3, and shall include at least the following positions:

(a) When the upper and lower booms are in a horizontal, or in their most horizontal, plane and extended to maximum reach
(b) When the lower boom is horizontal, or near horizontal, and to the side of the vehicle, and the upper boom is in the most vertical position possible
(c) When the lower boom is at maximum travel from the stowed position, and the upper boom is horizontal and 45 degrees to the side of the vehicle, or over the four corners of the vehicle. Rotate turntable with twice the specified load (see 3.6.5) 360 degrees and stop each 15 degrees. Reverse the turntable and stop each 20 degrees.

The inner dual tire shall be observed for conformance to the stability requirements of 3.3.3 with twice the load (see 3.6.5). The test shall be repeated with 1.5 times the specified payload (see 3.6.5) and the outriggers and all tires observed for conformance to the stability requirements of 3.3.3 with 1.5 times the load.

* 4.4.6 Wind test. With the vehicle not loaded, with the outriggers extended, and the rated capacity placed in the basket(s), the boom assembly shall be extended to maximum height and shall be rotated to that position whereby the greatest area of the basket(s) is parallel to the longitudinal centerline of the vehicle. The specified wind load, generated by artificial means or by an equivalent simulated load, shall be applied to the vehicle from either side (see 3.3.4).

4.4.7 Speed test. With the vehicle resting on approximately level ground, the basket(s) shall be loaded with specified load (see 3.6.5). The basket(s) shall be traversed and elevated to determine conformance to the requirements in 3.3.5, 3.3.6 and 3.3.7.

* 4.4.8 Creep test. With the vehicle longitudinal on a 5 percent grade, without leveling the vehicle, with specified load (see 3.6.5) in the basket(s), and with the hydraulic fluid stabilized at or near ambient temperature, the basket(s) shall be extended to the side with lower boom in vertical or nearest capable vertical position, and the upper boom in horizontal position. The position shall be held for at least 1 hour for maximum horizontal creep of 25 mm/h

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(1 inch per hour) and vertical settling of 50 mm/h (2 inches per hour) to determine conformance to requirements in 3.3.8.

4.4.9 Performance test. The basket(s) shall be subjected, but not limited, to the following tests:

(a) Check to determine whether boom assembly can be elevated without outriggers
(b) Determine if outriggers have locked automatically when in an extended position and in a retracted position
(c) Operate equipment using each set of controls and emergency controls separately
(d) Check to insure that the lower controls override the basket controls
(e) Check to insure that the entire system can be operated smoothly and without jerk motions in any position
* (f) Check to determine whether the basket(s) remains level in all positions with the specified loads in the basket(s)
(g) Check to determine whether the basket(s) is safely lowered from maximum reach position, with rated payload on the basket(s) floor, to within 910 mm (36 inches) above the ground, with the hydraulic pump and engine not operating, except as specified in 3.3.6
(h) Check the remote control starter and cutoff on basket(s) to insure starting and stopping the engine with a specified payload on the basket(s)
* (i) Check overload capabilities by attempting to lift from rest position a load that will result in exceeding the manufacturer's recommended hydraulic system pressure setting (see 3.6.4.4)
(j) Check the hydraulic system, including the pump, at 10 percent above the pump manufacturer's maximum operating pressure setting for normal workload conditions
(k) Check operation of hydraulic tool circuits with outriggers up and outriggers down.

* 4.4.10 Dielectric tests. The vehicle shall be tested as specified in the Qualification Test of ANSI A92.2, for Category A and B aerial devices for a 69,000 volts (69 kV) unit rating.

4.4.11 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.4.12 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken. In addition, subsequent vehicles that do not conform to the accepted production sample vehicle shall be rejected until corrected.

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* 4.5 Availability of ANSI A92.2. The contractor shall make available to the Government at the point of final acceptance and test, current copies of ANSI A92.2. The document shall be surrendered to the Government for the personal use of the Government representatives in conjunction with the contract for the duration of the contract.

4.6 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.4.23 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government for distribution lines not exceeding 69,000 volts with the basket liner installed for utility line maintenance, replacement of lamps, tree trimming, painting, window washing, and many other types of overhead construction and maintenance work.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, size, class and model of vehicle required (see 1.2, 1.2.3 and 6.5).
* (c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of appropriate service for marking (see 3.1.1.2).
(e) Concealed markings, if required (see 3.1.1.2).
(f) Rustproofing, if required (see 3.1.1.3).
(g) Tropical rustproofing, if required (see 3.1.1.3).
(h) Trailer towing package, if required (see 3.1.1.6).
* (i) Electric trailer brake controller, if required (see 3.1.1.8).
(j) Drive hubs, if required for model 4x4, classes A and B (see 3.1.1.10).
(k) Silicone brake fluid, if required and available for classes A and B (see 3.1.1.11).
(l) Extended horizontal reach, if required for sizes 75 through 100 (see 3.3.6.1).

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*	(m)	Satisfactory operation on JP-4, JP-5 and JP-8, if required (see 3.4.1.1).
	(n)	Gasoline engine, if required for model 4x2 vehicles (see 3.4.1.2).
	(o)	Power plant heaters and fuel warmer, if required (see 3.4.1.8).
*	(p)	Vernier throttle control, if required (see 3.4.1.9).
*	(q)	Starting motor circuit breaker, if required (see 3.4.2.1).
*	(r)	Alternator capacity, if other than as specified (see 3.4.2.3).
*	(s)	Two speed transfer case, model 4x4, if required (see 3.4.5.1.2).
*	(t)	Automatic transmission, if required (see 3.4.5.3).
	(u)	Two-speed axle, if required for model 4x2 and 6x4 vehicles (see 3.4.9.2).
	(v)	Traction control, if required (see 3.4.9.3).
	(w)	Wide base type tires and wheels in lieu of duals, if required (see 3.4.10).
	(x)	Disc type wheels, if required (see 3.4.10).
	(y)	Bias ply tires, if required (see 3.4.10.1).
*	(z)	Low profile tires, if required (see 3.4.10.1).
	(aa)	A spare tire carrier, if required (see 3.4.10.3).
	(ab)	Spare wheel or rim, if required (see 3.4.10.4).
	(ac)	Spare tire assembly, if required (see 3.4.10.5).
	(ad)	Full air brakes, if required for model 4x2, class A and B vehicles (see 3.4.11.2).
	(ae)	Trailer brake control system, if required (see 3.4.11.4).
	(af)	Brake controls for use from a towing vehicle, if required (see 3.4.11.5).
*	(ag)	Forward tilting hood and fender assembly, if required (see 3.4.12).
	(ah)	Individual driver and passenger seats, if required (see 3.4.12.1).
*	(ai)	Air ride driver's seat, if required (see 3.4.12.1).
	(aj)	Crew cab, if required (see 3.4.12.2).
*	(ak)	Intermittent windshield wipers, if required (see 3.4.14).
	(al)	Tools, if required (see 3.4.16.1).
*	(am)	Remote control curbside rearview mirror, if required (see 3.4.18.2).
*	(an)	Heated rearview mirrors, if required (see 3.4.18.2).
	(ao)	Air operated horn, in addition, if required (see 3.4.19).
	(ap)	Backup alarm, if required (see 3.4.21).
	(aq)	Front mounted winch, if required (see 3.4.22).
	(ar)	Cooling system protection down to -54°C (-65°F), if required (see 3.4.23.1).
	(as)	Water cask, if required (see 3.5.5.2).
	(at)	Portable gasoline engine driven generator, if required (see 3.5.5.3).
	(au)	Lifting attachment for conductors, if required (see 3.6.4.8).
	(av)	Non-insulated guardrail type basket, if required (see 3.6.5.2).
	(aw)	Rotating basket(s), if required (see 3.6.5.7).
	(ax)	Hydraulic fluid conforming to MIL-H-83282, if required (see 3.6.8.5.1).

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(ay) Rechargeable electrical system, if required in addition to PTO (see 3.6.9).
(az) Auxiliary engine, if required in addition to PTO (see 3.6.10).
(ba) Auxiliary engine fuel tank, if required (see 3.6.10.6).
(bb) Air compressor (furnished only with auxiliary engine), if required (see 3.6.10.9).
(bc) If a ground hydraulic tool outlet is not required (see 3.6.11.1).
(bd) If a basket(s) hydraulic tool outlet and hose are not required (see 3.6.11.1.1).
(be) Hydraulic capstan hoist, if required (see 3.6.11.3).
(bf) Hydraulic impact wrench and drill, if required (see 3.6.11.4).
(bg) Hydraulic chain saw, if required (see 3.6.11.5).
(bh) Hydraulic chain saw, pole mounted, if required (see 3.6.11.6).
(bi) Hydraulic circular saw, pole mounted, if required (see 3.6.11.7).
(bj) Hydraulic pruner, pole mounted, if required (see 3.6.11.8).
(bk) Hydraulic hose reel, if required (see 3.6.11.11).

6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contract. The SAE Work Sheet Item 1 should include vehicle model number, engine model number, and vehicle type, class. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.613
Table 4	- Area Factor. 0.173
Table 5	- Velocity Factor. 250.0
Table 6	- Altitude Factor. 1.00
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, tabulated 454 kg (1,000 pounds)) and the engine rpm (to nearest 100 revolutions) which is required for 80 km/h (50 mph) geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade Factor. 0.75
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0.

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* 6.4 Basket insulation. The 30,000 volt basket insulation specified in 3.6.3.3 is required to obtain a heavy duty basket devoid of thin areas and imperfections in the basket walls.

6.5 Cautionary note. Requisitioners are cautioned that over-the-road maneuverability of size 120 vehicle may be greatly restricted by the length of the booms when the booms are in the stowed position.

* 6.6 Subject term (key word) listing.

Aerial basket
Basket leveling system
Fiberglass basket
Horizontal reach
Hydraulic system
Outrigger
Rotating turntable
Stabilizing equipment
Upsetting stability.

* 6.7 Identification of changes. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AT
Navy - YD
Air Force - 99

Preparing Activity:
Army - AT

Project No. 2320-0575

Review activities:
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